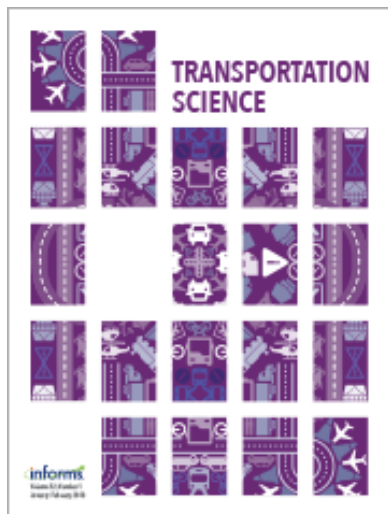




Transportation Science

Publication details, including instructions for authors and subscription information:
<http://pubsonline.informs.org>

Cost Allocation for Less-Than-Truckload Collaboration via Shipper Consortium

Minghui Lai, Xiaoqiang Cai, Nicholas G. Hall

To cite this article:

Minghui Lai, Xiaoqiang Cai, Nicholas G. Hall (2022) Cost Allocation for Less-Than-Truckload Collaboration via Shipper Consortium. *Transportation Science* 56(3):585-611. <https://doi.org/10.1287/trsc.2021.1066>

Full terms and conditions of use: <https://pubsonline.informs.org/Publications/Librarians-Portal/PubsOnLine-Terms-and-Conditions>

This article may be used only for the purposes of research, teaching, and/or private study. Commercial use or systematic downloading (by robots or other automatic processes) is prohibited without explicit Publisher approval, unless otherwise noted. For more information, contact permissions@informs.org.

The Publisher does not warrant or guarantee the article's accuracy, completeness, merchantability, fitness for a particular purpose, or non-infringement. Descriptions of, or references to, products or publications, or inclusion of an advertisement in this article, neither constitutes nor implies a guarantee, endorsement, or support of claims made of that product, publication, or service.

Copyright © 2021, INFORMS

Please scroll down for article—it is on subsequent pages



With 12,500 members from nearly 90 countries, INFORMS is the largest international association of operations research (O.R.) and analytics professionals and students. INFORMS provides unique networking and learning opportunities for individual professionals, and organizations of all types and sizes, to better understand and use O.R. and analytics tools and methods to transform strategic visions and achieve better outcomes.

For more information on INFORMS, its publications, membership, or meetings visit <http://www.informs.org>

Cost Allocation for Less-Than-Truckload Collaboration via Shipper Consortium

Minghui Lai,^a Xiaoqiang Cai,^{b,c,*} Nicholas G. Hall^d
^aSchool of Economics and Management, Southeast University, Nanjing 211100, People's Republic of China; ^bThe Chinese University of Hong Kong-Shenzhen, 518172 Shenzhen, People's Republic of China; ^cThe Shenzhen Research Institute of Big Data, 518172 Shenzhen, People's Republic of China; ^dFisher College of Business, The Ohio State University, Columbus, Ohio 43210

*Corresponding author

Contact: laimh@seu.edu.cn,  <https://orcid.org/0000-0002-7381-0704> (ML); xqcai@cuhk.edu.cn,  <https://orcid.org/0000-0001-9829-5246> (XC); hall.33@osu.edu,  <https://orcid.org/0000-0003-4484-9252> (NGH)

Received: December 23, 2019

Revised: July 25, 2020; January 17, 2021

Accepted: April 4, 2021

Published Online in Articles in Advance:
November 4, 2021

<https://doi.org/10.1287/trsc.2021.1066>

Copyright: © 2021 INFORMS

Abstract. We study the problem of collaborative less-than-truckload (LTL) transportation in the form of a shipper consortium, which is operated by a third-party logistics provider (3PL) through a cross-dock/pooling network. The 3PL has responsibility for planning the combined loads prior to actual shipments, hiring and routing carriers to execute shipping, and allocating the cost to the shippers in the consortium. Shippers receive substantial cost savings from combined truckload shipments. However, achieving consolidation and realizing this benefit requires addressing two essential issues: (i) how to find an approximately optimal consolidation solution in a large network with many freights and (ii) how to determine a fair cost allocation rule among the shippers' consolidated freights that ensures budget balance while minimally violating coalitional stability. Our work resolves these two issues. We formulate a time-space network flow model of the problem under both incremental and all-unit discount structures of LTL rates and propose a computationally efficient algorithm based on local search heuristics. We model the problem of allocating cost to the shippers as a cooperative game. We decompose and linearize the Lagrangian dual problem by using total unimodularity and concavity. We propose an efficiently computable cost allocation rule from the linearized dual models. The dual rule ensures stable cooperation but may have underallocation equal to a duality gap. To cover the underallocation, we further develop a budget covering procedure and define an ϵ -core allocation with desirable properties. Through extensive computational experiments, we find that the shipper consortium reduces total shipping costs by more than 40% in most cases; meanwhile, the ϵ -core allocation is typically in the core for small-scale networks while violating stability by at most 5% for large-scale networks and provides consolidated freights with more than 50% individual cost savings on average.

History: This paper has been accepted for the *Transportation Science* Special Issue on Transportation in the Sharing Economy.

Funding: This work was supported by the National Natural Science Foundation of China [Grants 71971057, 71531010, and 71732003]. This research is also supported in part by the Leading Talent Program of Guangdong Province [Grant 2016LJ06D703] to X. Cai and by the Berry Professorship at the Fisher College of Business, The Ohio State University, to N. G. Hall.

Supplemental Material: The online appendix is available at <https://doi.org/10.1287/trsc.2021.1066>.

Keywords: collaborative logistics • sharing economy • less-than-truckload shipping • cooperative game

1. Introduction

In less-than-truckload (LTL) shipping, a core principle behind shipping costs is that as volume goes up, per unit shipping costs go down. With the support of advanced technologies (e.g., Internet of Things, cloud computing, robotics, and automation), it is a growing trend in the logistics industry that shippers set up partnerships with each other to reduce freight costs. Robinson (2016) summarizes five common forms of shipper partnerships—mode optimization, aggregation, multistop truckload, pooling distribution, and shipper consortium—ordered from the easiest to

implement to most difficult. We study shipper consortium in this paper.

According to Robinson (2016), in a shipper consortium, several companies collaborate to combine their freights into larger shipments and shift from LTL to truckload (TL) whenever possible, with the ultimate goal of decreasing handling and costs. As in a case study by Creemers et al. (2017), a *shipper consortium is differentiated from traditional LTL shipping in that shippers can actively synchronize the shipments even by re-scheduling the timing of their individual freights, and the combined loads are planned prior to actual shipments*

executed by carriers. The flexibility on timing allows for more potential cost savings. It is particularly beneficial for those companies that have small-size freights with overlapping shipping lanes and delivery time windows and agree to participate in a consortium as part of a long-term strategy (Robinson 2016). The freights are typically regular LTL shipments weighing between 2,000 and 5,000 lbs or many minimum-charge shipments (i.e., less than 500 lbs).

Because of implementation complications, it is most common that a shipper consortium is organized by a third-party logistics provider (3PL) that operates a network of cross-dock and pools (Robinson 2014). The 3PL has the responsibility of hiring and routing carriers, tracking freights, handling freight bill payment and audit, and allocating the cost to the shippers in the consortium. Typically, a freight may be first sent to a cross-dock, consolidated with other freights into a TL shipment, from where it is transported to a pool, where it is de-consolidated and finally delivered to the destination via LTL. These small freights thereby can bypass a relatively expensive long-haul LTL network.

Some leading logistics providers have already established shipper consortium programs. One prominent example is the TranSaver Program of CT Logistics. CT Logistics can pool and route all participating shippers' freights, both outbound and inbound, to take advantage of the same competitive motor and air carrier pricing that only larger shippers could receive. TranSaver generates an average discount on prepaid outbound shipments of up to 58% for interstate and 56% for intrastate and has saved shippers more than \$2 billion dollars in the last 30 years (CT Logistics 2017). Another example is the Retailer Consolidation Program operated by CaseStack since 2003. By working closely with leading retailers, CaseStack combines suppliers' weekly LTL purchase orders through the network of consolidation centers or via a cross-dock point and offers proportional full truckload pricing instead of costly LTL rates to the suppliers (Inbound Logistics 2016). Participating suppliers enjoy savings of 15%–50% over prepaid LTL while complying with retailers' delivery time windows (CaseStack 2015).

Although some consortia have worked well for decades, the operation of a shipper consortium has proved to be very challenging, and many other consortia have disbanded. Robinson (2016, p. 9) points out that "unless every participant is 'all in' and willing to focus on the consortium as part of a long-term strategy, disagreements can arise on which shipper is contributing what, what the pricing is, or how savings are split." Effective cooperation via shipper consortium requires resolution of these important practical issues.

In this paper we study the operational planning and fair cost allocation problem for the shipper

consortium organized by a 3PL. Specifically, we consider the following process of cooperation.

1. A group of *shippers* join a shipper consortium coordinated and implemented by a 3PL.

2. Based on the estimated shipment demands of the shippers over a given period (e.g., one year), the 3PL on behalf of the shippers negotiates with the carriers for better LTL rates as well as TL rates in advance.

3. The 3PL prespecifies a number of time periods (e.g., three to five days or one week) as the planning horizon. At the start of each planning horizon, *shippers* in the consortium submit their freights to the 3PL that must be delivered by the end of the planning horizon.

4. The 3PL optimally selects a subset of these submitted freights to go through the cross-dock/pooling network and plans the consolidation of freights in the network in order to maximize total shipping cost savings.

5. The 3PL hires *carrier(s)* to deliver the combined shipments through the network, and the shipping costs are charged to the shippers based on the established contracts; the 3PL then distributes the total cost to each individual freight according to an allocation rule proposed in the paper. We consider two types of LTL cost structure, that is, incremental discount and all-unit discount, which are both common in the literature and practice.

The contributions of this paper are summarized here.

- i. Shippers' collaboration is a significant problem in the LTL logistics industry (Terry 2015, Robinson 2016, Arslan et al. 2020). We propose a new application of cooperative game theory to collaborative logistics via shipper consortium that incorporates various practical features.

- ii. The shippers' cost savings from collaboration depend on how the 3PL optimizes the consolidation of shipments. However, this optimization problem is highly intractable. We develop an approximation algorithm based on local search heuristics. Extensive computational experiments show that this algorithm can substantially reduce shipping cost by more than 40% compared with the standalone cost and also solves most small-scale instances to optimality quickly.

- iii. A fair split of the shipping cost is critical for the success of a shipper consortium. We first decompose and linearize the Lagrangian dual problem with piecewise linear concave cost structures to determine a coalitional stable rule that may not cover the total cost. Then, we develop a budget covering procedure and define an ϵ -core cost allocation with desirable properties. In contrast to many linear programming based methods for approximate core allocation in the literature, this Lagrangian dual based method is able to solve very large-scale problems without becoming intractable. Extensive computational experiments show that the ϵ -core allocation only results in relatively small deviations from stability, mostly zero (i.e., in the core) for small-scale networks while typically no more than 5% for large-scale networks.

iv. We conduct extensive computational experiments based on practical cost structures to understand the properties of our proposed algorithms for managing shipper consortia. We find that the service density and freight density in the network, the capacity competition level, and freights' delay costs have the most significant effects on the system cost savings and cost allocation stability. Moreover, our algorithms work better under the incremental discount structure.

The remainder of this paper is organized as follows. In Section 2, we review the relevant literature on LTL collaborative logistics and network flow cooperative games and point out the main differences of our paper. In Section 3, we formally define the centralized optimization problem of shipper consortium. In Section 4, we solve the centralized optimization problem and a Lagrangian dual problem. In Section 5, we study the shippers' collaboration as a cooperative game. Computational studies are reported in Section 6. Section 7 provides a summary of our results, important insights about our proposed algorithms for managing shipper consortia, and several suggestions for future research. All supplemental materials and proofs appear in the online appendix.

2. Literature Review

Collaborative logistics has become increasingly important and the related literature has dramatically grown during the last decade. Recent surveys have extensively reviewed state of the art in collaborative logistics theories and practices (Guajardo and Rönnqvist 2016, Gansterer and Hartl 2018, Cleophas et al. 2019, Pan et al. 2019, Ferrell et al. 2020, Gansterer and Hartl 2020), summarizing research advances and gaps. In particular, Guajardo and Rönnqvist (2016) review cooperative game cost allocation methods and classify them into two types: *traditional methods* based on well-established concepts, for example, Shapley value, nucleolus, proportional rule, and dual rule; and *ad hoc methods* dependent on problem context, for example, core and ϵ -core. As these recent reviews are already up to date, in the following, we only survey the research on shippers' LTL collaboration (Section 2.1) and capacitated network flow cooperative games (Section 2.2), which are most relevant to our study. We highlight the cost allocation methods in the literature.

2.1. Shipper Cooperation

A majority of the studies in this literature investigate the collaboration of carriers, whereas shippers' collaboration is rarely considered (Gansterer and Hartl 2020). Yet, early studies on basic vehicle routing games and the recent variants are also relevant, where the central issue is to allocate the joint cost among the

served customers. Göthe-Lundgren, Jörnsten, and Värbrand (1996) and Engevall, Göthe-Lundgren, and Värbrand (2004) consider the core, nucleolus, and Shapley value for cost allocation in vehicle routing games. Özener, Ergun, and Savelsbergh (2013) extend their game models and propose a deterministic inventory routing game. Proportional rule and per-route-based and duality-based ϵ -core rules are developed for cost allocation. Osicka, Guajardo, and van Oost (2020) propose a cooperative location-routing game and study several variants of it. The Shapley value, nucleolus, and the lexicographic equal profit method for cost allocation are tested on small-scale random instances.

Later studies on shipper collaboration investigate more general LTL transportation problems. We first review the centralized planning studies and then the cost allocation game studies. Chabot et al. (2018) study the planning for *multistop truckload* or *bundle shipping* collaboration, where shippers synchronize their shipments to share a single line-haul. Arslan et al. (2020) propose a shippers' collaborative strategic network design problem, where the flow costs on arcs have an incremental discount structure. *This is perhaps the first centralized planning study on shipper consortium problem. However, they only study the optimization algorithms for centralized planning.*

Because of the complicated nonconvex cost structures for shipper collaboration problems, the associated cooperative games generally have an empty core. Thus, the literature typically uses traditional allocation methods, especially the Shapley value and proportional rules, and may further limit the size of coalition. In the cost allocation problem for shippers' cooperative bundle shipping: Vanovermeire et al. (2014) and Vanovermeire and Sörensen (2014a) both use only the Shapley value; meanwhile, in the special case of two-shipper coalition, Vanovermeire and Sörensen (2014b) compare the Shapley value, nucleolus, and the equal-charge method, and Tinoco, Creemers, and Boule (2017) compare three proportional rules. Nguyen, Dessouky, and Toriello (2014) study a volume-based proportional rule in a perishable LTL freight consolidation game by shippers with the same break-bulk destination. Similarly, Ramaekers et al. (2017) study a shipment aggregation cooperative game in multimodal barge transport and select the proportional rule, Shapley value, and equal-profit methods for cost allocation. An exception that develops ad hoc methods is Yilmaz and Savasaneril (2012), which constructs a linear program to find an allocation that only satisfies the budget balance and individual rationality conditions.

Because of emptiness of the core in general, some studies investigate the coalition structure or coalition formation (Kaewpuang et al. 2017, Guajardo et al.

2018). On the other hand, only a few studies apply the incentive mechanism design methodology to shipper collaboration. Zhang et al. (2018) design a truthful Moulin mechanism for shippers' cost sharing in a freight consolidation system with one consolidation center and a common destination while only approximately guaranteeing budget balance.

Gansterer and Hartl (2020) point out that the huge majority of studies do not contribute new ideas to collaborative transportation but apply well-established allocation methods (mostly the Shapley value). However, any Shapley-type or nucleolus-type cost allocations would be extremely time-consuming to compute (Özener and Ergun 2008), and the Shapley value may not be stable (Shapley 1971) or individually rational (Vanovermeire and Sörensen 2014b). We contribute to this field by developing ϵ -core allocations based on easily computable Lagrangian dual, in contrast to Özener, Ergun, and Savelsbergh (2013), who use linear relaxation of set-partitioning/covering models. Their models require generating sufficiently many routes, which is not easy for large-scale problems. Moreover, the shippers' collaborative bundle shipping problem, where LTL freights are consolidated through multistop truckloads without any intermediate transfer, is significantly different from the shipper consortium problem studied in our paper.

2.2. Capacitated Network Flow Games

In the capacitated network flow cooperative games (CNFG), the participants pool their own resources and share the pooled capacities. This stream of literature can be classified as node-based and arc-based games depending on the type of capacity.

Node-based CNFG are typically studied in the context of Internet routing, where the participants offer their communicating capabilities (bandwidth) at nodes to broadcast contents across the network. If the coalition value is given by a convex optimization problem, Markakis and Saberi (2005) prove that the core is nonempty. In the more general nonconvex case, Altan and Özener (2019) develop ϵ -core allocations by using dual-based and partition-based rules derived from linear programming relaxation models.

In arc-based CNFG, participants collaborate by exchanging capacities on the network arcs to improve their profits. Agarwal and Ergun (2008) propose a linear multicommodity flow game for service network alliances and show that the dual rule and a capacity exchange mechanism by inverse optimization are both in the core. In the case of liner shipping alliances, with fixed charges on capacities, the core is generally empty, and hence Agarwal and Ergun (2010) restrictively assume fixed routes and design a capacity exchange mechanism by using inverse optimization techniques. Zheng et al. (2015) extend their

mechanism to a more practical liner shipping collaboration case. In the case of air freight alliances, Houghtalen, Ergun, and Sokol (2011) similarly apply inverse optimization methods to define a capacity exchange mechanism in the core under the condition of limited control behavior and linear cost structure.

The CNFG and our cooperative game are different in nature. The incentive for participating in a CNFG is sharing capacities by pooling. This is clearly not the case in our problem, where the capacities are exogenously given and incentives are mainly from cost savings by TL rates. Moreover, the ϵ -core allocations of node-based CNFG in Altan and Özener (2019) are derived by scaling up the dual rules and require generating sufficiently many partitioning sets, whereas individual rationality and participants' actual contributions to the system are ignored; the capacity exchange mechanisms of arc-based CNFG require restricting participant's behavior. In our problem, the participation is fully voluntary without any restriction. Thus, none of these allocation methods are practically applicable to our problem.

3. The Shipper Consortium Collaboration Problem

Throughout this paper, we generally represent a scalar by a lowercase letter, for example, x ; a vector by a bold lowercase letter, for example, $\mathbf{x}$; a matrix by an uppercase letter, for example, A ; and a set by a mathematical calligraphic upper case letter, for example, $\mathcal{A}$. In addition, $x^+ := \max\{x, 0\}$ and $x^- := \min\{x, 0\}$. The notations for the main parameters and variables are summarized in Table 1.

We consider the collaboration problem in a given planning horizon of T units of time. For ease of modeling, we assume that the time has been discretized as $\{1, 2, \dots, T\}$. The length of each time period could be one day or an integral number of hours, depending on the required granularity of operational planning.

Let $\mathcal{N}$ denote the collaborative shipper consortium organized by a 3PL, consisting of a number of shippers. The set of LTL freights within the planning horizon submitted by the shippers is denoted by $\mathcal{S}$. Each freight $s \in \mathcal{S}$ is specified by an origin o_s , a destination d_s , weight w_s (unit: lbs), an available time t_s^o , and a due time t_s^d . The 3PL selects a subset of these freights into the cross-docking/pooling (CD-PL) network to maximize the total shipping cost savings, because it may not be economical to send all LTL freights via consolidation. If s is not selected into the network, its stand-alone cost is denoted by R_s , corresponding to direct shipping.

The 3PL plans to consolidate the LTL freights into TL shipments whenever possible through its CD-PL network. In the following, we first describe the

Table 1. Summary of Notations for Parameters and Variables

	Descriptions
Parameters	
s	A freight specified by origin o_s , destination d_s , weight w_s , available time t_s^o , and due time t_s^d
$(\mathcal{V}, \mathcal{A})$	The transportation network with node set $\mathcal{V}$ and arc set $\mathcal{A}$
$(\tilde{\mathcal{V}}, \tilde{\mathcal{A}})$	The time-space network with vertex set $\tilde{\mathcal{V}}$ and arc set $\tilde{\mathcal{A}}$
$\mathcal{I}(v)$	The set of incoming arcs of vertex v
$\mathcal{O}(v)$	The set of outgoing arcs of vertex v
$\tilde{\mathcal{A}}_{\text{pooling}}^c$	The set of pooling arcs in the time-space network
$\tilde{\mathcal{A}}_{\text{service}}^c$	The set of local arcs in the time-space network
$\hat{d}_s$	The sink vertex of freight s in the time-space network
q_v^s	The netflow of freight s at vertex v
U_e	The capacity of vehicle on arc e
f_e	The flat truckload rate on arc e
$c_e(x)$	The LTL rate function on arc e
$r_e^l(\tilde{r}_e^l)$	The (approximate) slope of segment l in a LTL rate function
$b_e^l(\tilde{b}_e^l)$	The (approximate) intercept of segment l in a LTL rate function
K_e	The number of segments in a LTL rate function
$R_s(\tilde{R}_s)$	The direct shipping cost (per lbs) of freight s
$h_e^s(\tilde{h}_e^s)$	The holding cost (per lbs) of freight s on arc e
$u_e^s(\tilde{u}_e^s)$	The lateness penalty cost (per lbs) of freight s on arc e
$\Gamma(\cdot)$	The system cost function
$\Psi(\cdot)$	The dual value function
$\Psi_e(\cdot)$	The dual value function on arc e
$\mu_s^D(\mu_e^L)$	The Lagrangian dual (Lagrangian-based) cost allocation to s
Variables	
x_e^s	The binary variable indicating freight s routed through arc e
y_e	The binary variable indicating a TL sent through arc e
λ_v^s	The dual variable for freight s visiting vertex v
γ_e	The dual variable for capacity constraint on arc e
π_e^s	The dual variable for allocating cost on arc e
δ_e^s	The dual variable for allocating fixed cost on pooling arc e
$\delta_{e,l}^s$	The dual variable for allocating fixed cost on local arc e

transportation network (Section 3.1) and the shipping cost structure (Section 3.2), then expand the transportation network into a time-space network (Section 3.3), and finally define the centralized optimization model (Section 3.4).

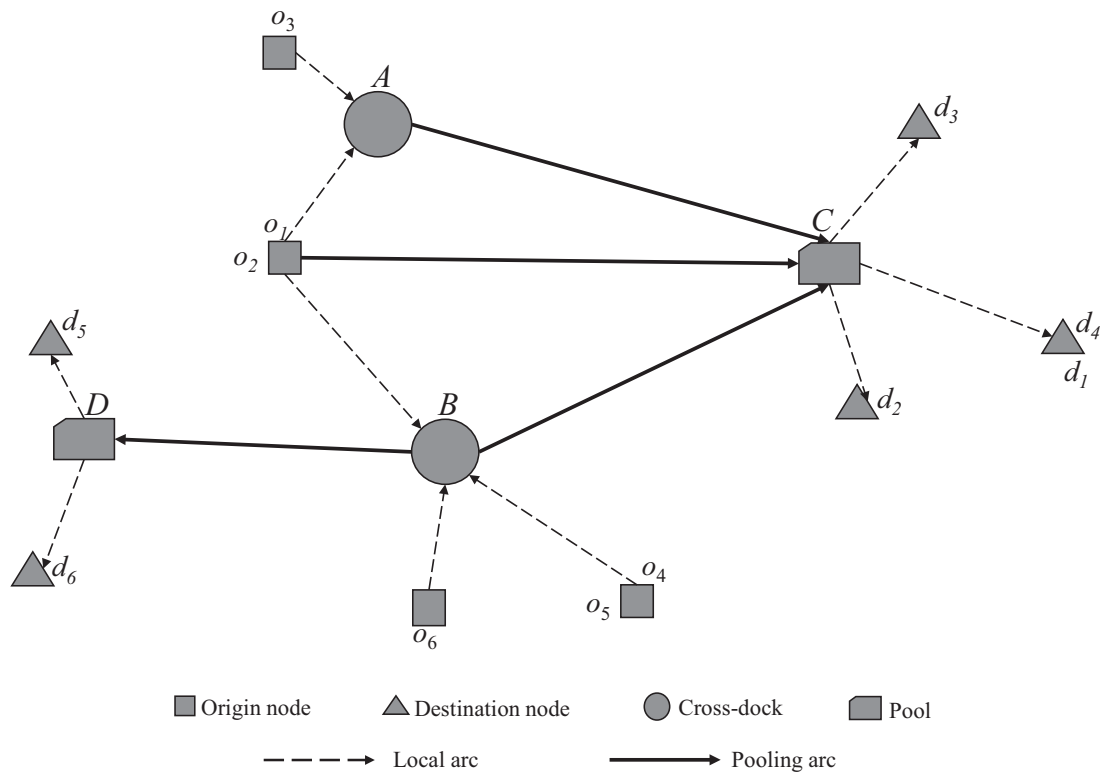
3.1. The Transportation Network

The CD-PL network commonly has a hub-and-spoke structure. We follow the recommendations by Caleres (2007) and iGPS (2019) to define this transportation network. Cross-docks function as distribution centers, where inbound shipments are transferred for further distribution. Pools act as de-consolidation centers for shipments with destinations in the region associated with the pool, and are typically close to customers. Therefore, in this network, each destination point is normally served by a *unique* pool, whereas each origin point may be assigned to more than one cross-dock.

We represent the CD-PL network by a directed graph $(\mathcal{V}, \mathcal{A})$, where $\mathcal{V}$ is the set of nodes, and $\mathcal{A}$ is the set of arcs linking the nodes. The node set $\mathcal{V}$ consists of the freights' origins and destinations, cross-docks, and pools. The set $\mathcal{A}$ is constructed to include the following directed arcs: (i) (o_s, v_c) linking each origin node o_s to each assigned cross-dock v_c , and (v_p, d_s) linking each destination node d_s from the

uniquely assigned pool v_p ; (ii) (v_c, v_p) connecting each pair of cross-dock v_c and pool v_p , if there is any potential flow; and (iii) (o_s, v_p) linking an origin node to a pool, if it is potentially economical to send a flow. Arcs of type (i) are referred to as *local arcs*, whereas types (ii) and (iii) are referred to as *pooling arcs*. Each arc e is associated with a travel time a_e , which includes both in-vehicle transit time and loading/unloading time.

We describe the shipment consolidation process organized by the 3PL according to Caleres (2007) and Robinson (2014). A freight if selected can be first sent to a cross-dock where it is unloaded from the inbound vehicle and reloaded onto an outbound vehicle together with other pending freights for further distribution, in order to build TL shipments if possible. The outbound shipment then travels to the pool where the combined freights are segregated and sorted by the destination and then reloaded onto local trucks for final delivery via LTL. In the case where the freights from the same origin shipped to the destinations within the service area of a pool are sufficiently large, such freights can be immediately sent to the pool via TL and then are separated for final delivery via LTL. Clearly, the route for each freight can visit at most one cross-dock/pool.

Figure 1. Example of CD-PL Transportation Network

A small example illustrating the transportation network is given in Figure 1, where there are six freights, two cross-docks, and two pools. Some freights share either the same origin or destination. In a consolidation scenario, freights s_1 and s_2 can be sent to cross-dock A for the consolidation with s_3 , or they can travel directly to pool C together.

For each freight in the network $(\mathcal{V}, \mathcal{A})$, the whole route may break into three segments (i.e., transfer from origin to cross-dock, long haul from cross-dock to pool, and transfer from pool to destination) or two segments (i.e., long haul from origin to pool and transfer from pool to destination). All segments of transportation are served by the carriers contracted by the 3PL on behalf of the shippers. Generally, in practice, different carriers may be hired to deliver the three segments, each offering competitive freight rates in their respective local market.

Although cost savings would be further improved, we do not allow shipment splitting, because it would lead to undesirable difficulties in tracking and managing them (Li, Bookbinder, and Elhedhli 2012) and is not common in the LTL industry.

Assumption 1. Each shipper can hold his freight on ground at the origin waiting for later pickup and also allows delivery later than the due time, although late delivery

incurs a penalty cost. The delivery of all freights is nonsplittable.

In contrast, the cross-docks and pools never hold inventory. Inbound shipments are moved directly from the inbound dock to the outbound dock, without the putaway and picking processes, and are shipped out in the same time period to achieve fast transit.

3.2. The Shipping Cost

The transportation mode of freights on local arcs are LTL, as only a few freights would be consolidated together on such arcs. On the other hand, the transportation mode on pooling arcs are generally required to be TL (Robinson 2014); otherwise, it would not be economical to operate the cross-docking/pooling infrastructure.

For each pooling arc e , let U_e denote the capacity of the vehicle used for transportation on pooling arc e , that is, the maximum weight of a truckload; the transportation cost for each TL is a flat rate denoted by f_e .

For each local arc e , the transportation cost function is an LTL rate. We consider two types of cost structures, that is, the incremental discount and the (modified) all-unit discount. Both are well documented in the literature (Russell and Krajewski 1992, Diaby and Martel 1993, Croxton, Gendron, and Magnanti 2003).

Assumption 2.

i. In the incremental discount case, for each local arc e , the transportation cost dependent on the total flow x is given by function $c_e(x)$:

$$c_e(x) = r_e^l x + b_e^l, \quad \text{if } Q_e^{l-1} \leq x < Q_e^l \text{ and } 1 \leq l \leq K, \quad (1)$$

where $1 \leq l \leq K$ is a segment of LTL rate, $r_e^1 \geq r_e^2 \geq \dots \geq r_e^K \geq 0$, $0 = Q_e^0 \leq Q_e^1 \leq \dots \leq Q_e^K$, $b_e^1 = 0$, and $b_e^l = \sum_{k=1}^{l-1} (r_e^k - r_e^{k+1})Q_e^k$ for $l \geq 2$. Equivalently, the cost function satisfies the piecewise linear concave property:

$$c_e(x) = \min_{1 \leq l \leq K} \{r_e^l x + b_e^l\}, \quad \forall x \geq 0. \quad (2)$$

ii. In the all-unit discount case, for each local arc e , the transportation cost dependent on the total flow x is given by function $c_e(x)$:

$$c_e(x) = \begin{cases} 0, & \text{if } x = 0; \\ r_e^l x + b_e^l, & \text{if } Q_{l-1} < x \leq Q_l, 1 \leq l \leq K. \end{cases} \quad (3)$$

where $1 \leq l \leq K$ is a segment of LTL rate, K is an even number, and $0 = Q_0 < Q_1 < \dots < Q_{K-1} < Q_K$. The slopes and intercepts satisfy: for increasing segments (k is even), $r_e^2 > r_e^4 > \dots > r_e^K > 0$, $b_e^2 = b_e^4 = \dots = b_e^K = 0$; for flat segments (k is odd), $r_e^1 = r_e^3 = \dots = r_e^{K-1} = 0$, $0 < b_e^1 < b_e^3 < \dots < b_e^{K-1}$; b_e^1 is the minimum charge rate. Q_l is the break point of increasing segment $l + 1$, where $l = 1, 3, \dots, K - 1$; Q_l is the indifference point between consecutive increasing segments l and $l + 2$, where $l = 2, 4, \dots, K - 2$.

By definition, $c_e(x)$ is a continuous and nondecreasing function. An example for each of the two LTL rate structures is illustrated in Figure 2. For most carriers, TL shipments normally range from 20,000 to 45,000 lbs, whereas LTL shipments are commonly less than

20,000 lbs (Lojistic 2019, Unishippers Global Logistics 2019). Then, we set $Q_e^K = 20,000$ and $Q_K = 20,000$ in the previous two cost structures. Moreover, the shippers' payment cost functions $c_e(x)$ only depend on their own shipment volume on the arc. This implies that it does not matter whether the carrier mixes non-member shippers' freights with theirs on the arc, which is standard in practical LTL pricing.

In addition, it is not cost free to synchronize shipments. We set h_s to be the holding cost per unit of time per pound for freight s to wait for pickup and u_s as the penalty cost per unit of time per pound for the late delivery of freight s .

3.3. The Time-Space Network

To model the shipments with different departure times, we construct a time-space network $(\tilde{\mathcal{V}}, \mathcal{A})$, where the vertex set $\tilde{\mathcal{V}}$ includes (v, t) for all $v \in \mathcal{V}$ and $t \in \{1, 2, \dots, T\}$. A pair of vertices (v_1, t_1) and (v_2, t_2) are connected with a directed service arc if arc $e = (v_1, v_2)$ exists in the physical transportation network $(\mathcal{V}, \mathcal{A})$ and $t_2 = \min\{1 \leq t \leq T : t \geq t_1 + a_e\}$. Each arc $[(v_1, t_1), (v_2, t_2)]$ represents a potential dispatch from v_1 at the end of period t_1 arriving at v_2 no later than the end of period t_2 . For each freight s , (o_s, t_s^0) is considered as the source vertex of the freight; we link each (o_s, t) to $(o_s, t + 1)$ with a directed ground arc for all $t \geq t_s^0$, representing that s is held at the origin for one time period; we also add a dummy vertex denoted by $\hat{d}_s$, which is the sink vertex for s , and link each (d_s, t) to $\hat{d}_s$ with a directed delivery arc to represent that s is delivered in period t , where $t \geq t_s^0 + 1$; finally, we add a directed dummy arc from (o_s, t_s^0) to $\hat{d}_s$, representing the direct shipping

Figure 2. Examples of Transportation Cost Functions

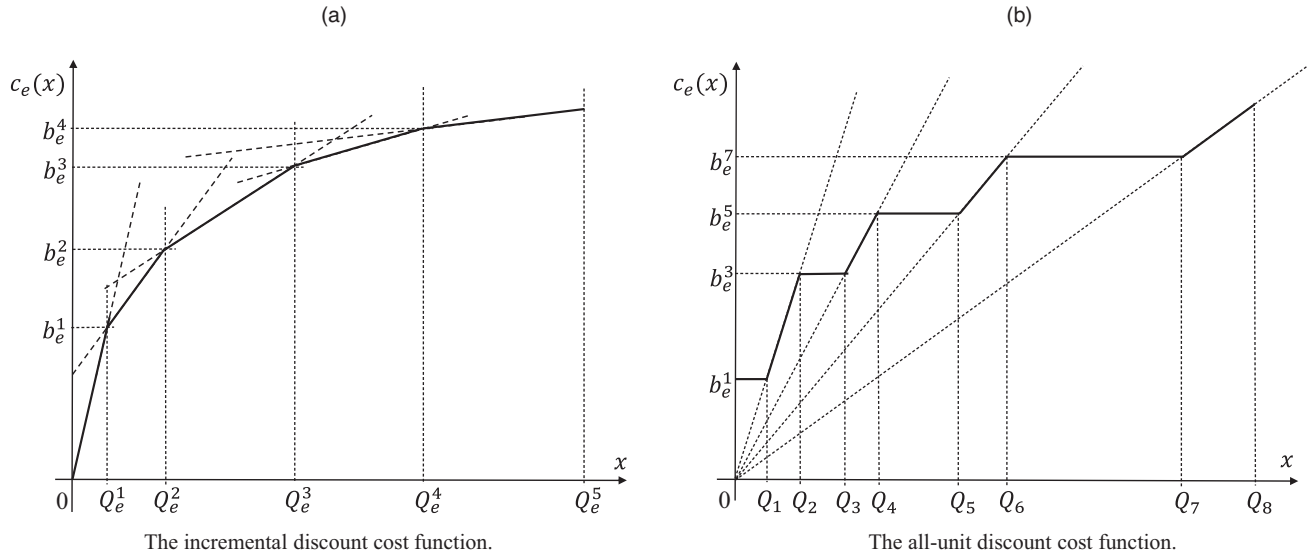
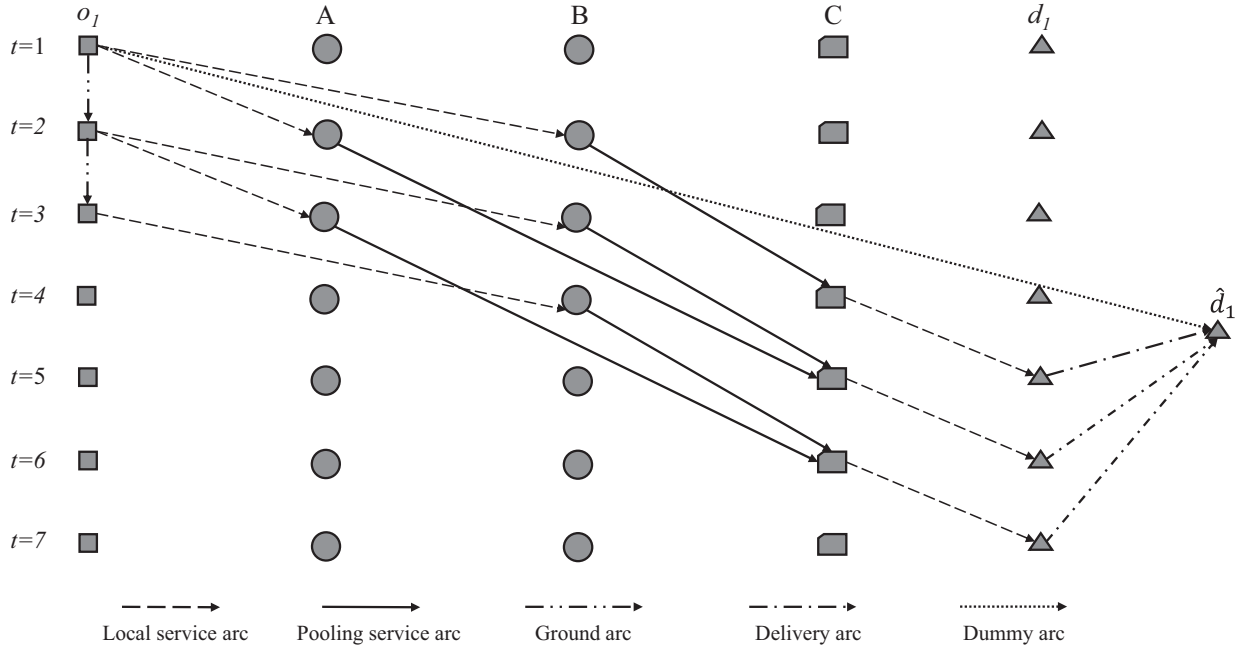


Figure 3. Illustration of the Time-Space Network

option. We only construct the arcs that would be on a feasible path from the source (o_s, t_s^o) to the sink $\hat{d}_s$ in the time-space network for each $s \in \mathcal{S}$. In Figure 3, we show the construction of vertices and arcs for freight $s_1 = (o_1, d_1, 1, 5, w_1)$ in Figure 1 as an example.

Let $\bar{\mathcal{A}}_{\text{service}}$, $\bar{\mathcal{A}}_{\text{ground}}$, $\bar{\mathcal{A}}_{\text{delivery}}$, and $\bar{\mathcal{A}}_{\text{dummy}}$ denote the sets of service arcs, ground arcs, delivery arcs, and dummy arcs in the time-space network, respectively. Then, $\bar{\mathcal{A}} = \bar{\mathcal{A}}_{\text{service}} \cup \bar{\mathcal{A}}_{\text{ground}} \cup \bar{\mathcal{A}}_{\text{delivery}} \cup \bar{\mathcal{A}}_{\text{dummy}}$. The time-space network is a directed acyclic graph. Let $\mathcal{I}(v)$ and $\mathcal{O}(v)$ denote the set of incoming and outgoing arcs of vertex v in the expanded network, respectively. The service arcs are further classified according to the type of the corresponding physical arcs, which are the set of pooling (service) arcs denoted by $\bar{\mathcal{A}}_{\text{service}}^C$ and the set of local (service) arcs denoted by $\bar{\mathcal{A}}_{\text{service}}^L$.

In this problem, a feasible *delivery path* of freight s is a path from (o_s, t_s^o) to $\hat{d}_s$ in the time-space network. Assume each freight has a feasible delivery path. The dummy vertex $\hat{d}_s$ is introduced merely for the convenience of representing the delivery paths, so that all paths of freight s have the same “ultimate destination.” It also becomes easier to represent a feasible path by the network flow balance condition. This construction is also used in Farvolden and Powell (1994).

Normally, in each period t , at most one TL could be possibly built on a pooling arc $e = (v_1, v_2)$. However, in some cases, it is possible to build more than one TL in time period t , we construct the necessary number of parallel arcs connecting (v_1, t) and (v_2, t') . Then, multiple

TLs are dispatched on physical arc e in the same time period is represented by a single TL flow on each of the parallel arcs. From now on, we assume that *all necessary parallel arcs have been constructed in the time-space network after analyzing the potential flows on each pooling arc*.

3.4. The Centralized Model

We formulate the shipper consortium’s collaborative transportation planning problem with a network flow model. Define binary variable $x_e^s = 1$ if freight s is routed through arc e and 0 otherwise, for all $s \in \mathcal{S}$ and $e \in \bar{\mathcal{A}}$. Define binary variable $y_e = 1$ if a TL is sent through arc e and 0 otherwise, for all $e \in \bar{\mathcal{A}}_{\text{service}}^C$. The total shipment flow through a service arc $e = [(v_1, t_1), (v_2, t_2)] \in \bar{\mathcal{A}}_{\text{service}}$, equal to $\sum_{s \in \mathcal{S}} w_s x_e^s$, is the load of the consolidated shipment executed by the carrier.

With a little abuse of notation, for each pooling arc $e = [(v_1, t_1), (v_2, t_2)] \in \bar{\mathcal{A}}_{\text{service}}^C$, the load must not exceed the capacity given by $U_e := U_{(v_1, v_2)}$ and the cost is the flat rate $f_e := f_{(v_1, v_2)}$. For each local arc $e = [(v_1, t_1), (v_2, t_2)] \in \bar{\mathcal{A}}_{\text{service}}^L$, the cost function is $c_e(\sum_{s \in \mathcal{S}} w_s x_e^s) := c_{(v_1, v_2)}(\sum_{s \in \mathcal{S}} w_s x_e^s)$.

The cost of a ground arc $e = [(o_s, t), (o_s, t+1)] \in \bar{\mathcal{A}}_{\text{ground}}$ for freight s is given by the holding cost $h_e^s = h_s \cdot w_s$. The cost of a delivery arc $e = [(d_s, t), \hat{d}_s] \in \bar{\mathcal{A}}_{\text{delivery}}$ for s is given by the lateness cost $u_e^s = u_s \cdot (t - t_s^d)^+ \cdot w_s$. The cost of a dummy arc $[(o_s, t_s^o), \hat{d}_s]$ is the direct shipping cost R_s . All the transportation costs, holding costs, and lateness costs are paid by the shippers themselves.

The objective is to determine the optimal flow $\mathbf{x} := (x_e^s)_{e \in \bar{A}, s \in S}$ and $\mathbf{y} := (y_e)_{e \in \bar{A}_{service}^C}$ that minimizes the total shipping cost.

$$\min \Gamma(\mathbf{x}, \mathbf{y}, S) := \sum_{e \in \bar{A}_{service}^C} f_e y_e + \sum_{e \in \bar{A}_{service}^L} c_e \left(\sum_{s \in S} w_s x_e^s \right) + \sum_{e \in \bar{A}_{ground}} \sum_{s \in S} h_e^s x_e^s + \sum_{e \in \bar{A}_{delivery}} \sum_{s \in S} u_e^s x_e^s + \sum_{e \in \bar{A}_{dummy}} \sum_{s \in S} R_s x_e^s \quad (4)$$

$$(CP) \text{ s.t. } \sum_{e \in \mathcal{I}(v)} x_e^s - \sum_{e \in \mathcal{O}(v)} x_e^s = q_v^s, \quad \forall v \in \bar{V}, s \in S; \quad (5)$$

$$\sum_{s \in S} w_s x_e^s \leq U_e y_e, \quad \forall e \in \bar{A}_{service}^C; \quad (6)$$

$$x_e^s \leq y_e, \quad \forall e \in \bar{A}_{service}^C, s \in S; \quad (7)$$

$$x_e^s, y_e \in \{0, 1\}, \quad \forall e \in \bar{A}, s \in S, \quad (8)$$

where q_v^s as defined as below is the demand for freight s at vertex $v \in \bar{V}$:

$$q_v^s = \begin{cases} -1, & \text{if } v = (o_s, t_s^0); \\ 1, & \text{if } v = d_s; \\ 0, & \text{otherwise.} \end{cases}$$

In the objective function $\Gamma(\mathbf{x}, \mathbf{y}, S)$, the five terms represent the total TL cost, the total LTL cost, the total ground holding cost, the total lateness cost and the total direct shipping cost, respectively. The objective is clearly equivalent to maximizing the total cost savings $\sum_{s \in S} R_s - \Gamma(\mathbf{x}, \mathbf{y}, S)$. Constraints (5) enforce network flow balance. Constraints (6) require that the TL shipment on each pooling arc cannot exceed the vehicle capacity limit. Constraints (7) require that a TL is sent along a pooling arc only if freight is routed through the arc. Constraints (8) state the binary requirement of flow variables to ensure no shipment splitting and at most one TL on each (parallel) pooling arc.

Remark 1. In the incremental discount case, if the LTL cost function $c_e(x)$ has a fixed charge, that is, $b_e^1 > 0$, we can add a new segment with a small range $[0, \bar{Q}_e^1]$, where $\bar{Q}_e^1 = \min\{\min_{s \in S} w_s, Q_e^1\}$, and this segment's slope and intercept are $\bar{r}_e^1 = (r_e^1 \bar{Q}_e^1 + b_e^1) / \bar{Q}_e^1$ and $\bar{b}_e^1 = 0$, respectively. The modified cost function is denoted by $\bar{c}_e(x)$. It is easy to check that $\bar{c}_e(x) = c_e(x)$ for all $x \geq \bar{Q}_e^1$ and $\bar{c}_e(0) = c_e(0) = 0$. Because shipment splitting is not allowed, the shipment flow on any local arcs is either 0 or no less than $\min_{s \in S} w_s \geq \bar{Q}_e^1$. Thus, model (CP) is still valid in such case. From now on, we implicitly assume that $b_e^1 = 0$ for all local arcs. $\square$

As all freights traveling through dummy arcs is a feasible solution, problem (CP) is always feasible.

Denote the optimal solution of problem (CP) by $(\mathbf{x}^*, \mathbf{y}^*)$ and the optimum cost by $\Gamma(S) := \Gamma(\mathbf{x}^*, \mathbf{y}^*, S)$.

4. Solution Algorithm

The variable-sized bin packing problem is known to be $\mathcal{NP}$ -hard, where NP stands for “nondeterministic polynomial time” (Friesen and Langston 1986). We establish that it is polynomially reducible to an instance of our problem.

Theorem 1. *The shipper consortium's centralized optimization problem (CP) is generally $\mathcal{NP}$ -hard.*

As a result of Theorem 1, we cannot expect to find an exact algorithm to solve problem (CP) in polynomial time.

In Section 4.1, we develop a primal approximation algorithm to find a promising local optimum solution, which gives an upper bound of the optimal cost. In Section 4.2, we solve the Lagrangian dual problem of (CP) by using a subgradient search to obtain a lower bound of the optimal cost. The difference between the primal optimum and Lagrangian dual optimum is the *duality gap*.

4.1. The Primal Approximation Algorithm

We propose a multistart local search heuristic (MSLS) that repetitively calls a local search function SPLS to search for a sequence of local optima and selects the best one as the final solution. Here, SPLS is a subroutine of heuristic MSLS. The pseudocodes of heuristic MSLS and the function SPLS are shown in Algorithms 1 and 2, respectively.

Algorithm 1 (Pseudocode of the Heuristic MSLS)

- 1: Construct an initial feasible solution $(\mathbf{x}^0, \mathbf{y}^0)$
- 2: Initialize the local optimum $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) = (\mathbf{x}^0, \mathbf{y}^0)$
- 3: **while** the stopping criterion is not satisfied **do**
- 4: Apply local search function SPLS with $(\mathbf{x}^0, \mathbf{y}^0)$ to find a new local optimum $(\mathbf{x}, \mathbf{y})$
- 5: **if** $\Gamma(\mathbf{x}, \mathbf{y}, S) < \Gamma(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}, S)$ **then**
- 6: Update $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \leftarrow (\mathbf{x}, \mathbf{y})$
- 7: **end if**
- 8: Randomly perturb $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ to generate a new initial feasible solution $(\mathbf{x}^0, \mathbf{y}^0)$
- 9: **end while**
- 10: **Return** the best local optimum $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$

The heuristic MSLS operates as follows.

i. **Initialization.** In lines 1 and 2, we sequentially assign a random delivery path to each freight while ensuring that the total flow on each pooling arc does not exceed the capacity. The direct shipping delivery path represented by the respective dummy arc is also feasible for each freight.

ii. **Local search.** In line 4, a shortest path local search function (SPLS) is repetitively called by the heuristic to improve a given initial solution $(\mathbf{x}^0, \mathbf{y}^0)$. The

pseudocode of this function is shown in Algorithm 2. In this subroutine, we search for a shortest path for one freight, while keeping the paths of other freights fixed; if we find a new solution that has lower total cost than the initial, we update the current solution and repeat the search process; otherwise, we consider a different freight; a local optimum is found if the solution cannot be improved any further, that is, $\Omega = \emptyset$.

Algorithm 2 (Pseudocode of Function **SPLS**)

```

1: function SPLS( $x^0, y^0$ )
2:   Set  $\Omega = \mathcal{S}$  and initialize the local optimum
   ( $x, y$ ) = ( $x^0, y^0$ )
3:   while  $\Omega \neq \emptyset$  do
4:     Randomly select a freight  $s \in \Omega$ 
5:     Fix the paths of other freights and solve a
     shortest path problem of freight  $s$ 
6:     Determine a new solution  $(\tilde{x}, \tilde{y})$  from the
     shortest path
7:     if  $\Gamma(\tilde{x}, \tilde{y}, \mathcal{S}) < \Gamma(x, y, \mathcal{S})$  then
8:       Update  $(x, y) \leftarrow (\tilde{x}, \tilde{y})$  and  $\Omega \leftarrow \mathcal{S}$ 
9:     else
10:      Set  $\Omega \leftarrow \Omega \setminus \{s\}$ 
11:    end if
12:  end while
13:  Return the local optimum  $(x, y)$ 
14: end function

```

Specifically, we construct the shortest path problem as follows. First, compute the incremental cost of arcs in the time-space network as:

$$\Delta c_e = \begin{cases} f_e, & \text{if } e \in \bar{\mathcal{A}}_{service}^C \text{ and } \sum_{s' \neq s} w_{s'} x_e^{s'} = 0; \\ 0, & \text{if } e \in \bar{\mathcal{A}}_{service}^C \text{ and } 0 < \sum_{s' \neq s} w_{s'} x_e^{s'} + w_s \leq U_e; \\ +\infty, & \text{if } e \in \bar{\mathcal{A}}_{service}^C \text{ and } \sum_{s' \neq s} w_{s'} x_e^{s'} + w_s > U_e; \\ c_e \left(\sum_{s' \neq s} w_{s'} x_e^{s'} + w_s \right), & \text{if } e \in \bar{\mathcal{A}}_{service}^L; \\ -c_e \left(\sum_{s' \neq s} w_{s'} x_e^{s'} \right), & \text{if } e \in \bar{\mathcal{A}}_{service}^L; \\ h_e^s, & \text{if } e \in \bar{\mathcal{A}}_{ground}; \\ u_e^s, & \text{if } e \in \bar{\mathcal{A}}_{delivery}; \\ R_s, & \text{if } e \in \bar{\mathcal{A}}_{dummy}. \end{cases} \quad (9)$$

Then, search for the shortest path from the source (o_s, t_s^0) to the sink $\hat{d}_s$ in the time-space network with arc cost Δc_e .

We set the new flow $\bar{x}_e^s$ of s corresponding to the shortest path solution while keeping $\bar{x}_e^{s'} = x_e^{s'}$ for all other s' , and then $\bar{y}_e = \max_{s \in \mathcal{S}} \bar{x}_e^s$ for all $e \in \bar{\mathcal{A}}_{service}^C$.

iii. **Random perturbation.** In line 8, given the currently best solution, we randomly choose n freights, where n is a predetermined number. While keeping the paths of nonselected freights unchanged, for each selected freight, we randomly generate a feasible delivery path without violating any capacity constraints.

iv. **Stopping criterion.** In lines 5–7, we keep recording the number of consecutively nonimproving iterations. If the search time reaches a predetermined limit denoted by $MAXP$ or the number of nonimproving iterations exceeds a predetermined number denoted by NUM , stop; otherwise, restart search with the new initial solution derived from random perturbation.

Clearly, **MSLS** is a single-trajectory search algorithm. The performance depends on the specification of parameters $MAXP$, NUM and n , which are empirically determined. As for the perturbation size n , we may use a large value in the early iterations, diversifying the search to avoid getting trapped in a local optimum, and then use a small value later for intensifying the search in a promising solution space. The output $(\tilde{x}, \tilde{y})$ of algorithm **MSLS** provides an approximation to the exact optimum with system cost $\bar{\Gamma}(\mathcal{S}) := \Gamma(\tilde{x}, \tilde{y}, \mathcal{S})$.

We justify the use of algorithm **MSLS** for the shipper consortium problem.

Remark 2. As in the literature (Croxtan, Gendron, and Magnanti 2003, Gendron and Gouveia 2017, Arslan et al. 2020), we may reformulate problem (CP) as a multiple-choice integer programming (MCIP) model by introducing a set of binary variables for each segment of the cost function. A branch-and-bound exact algorithm with strengthened valid inequalities was developed by Gendron and Gouveia (2017) and tested on random networks with size up to 25 nodes, 150 arcs, and 100 commodities. However, our model is based on a time-space network. Even for a small-scale transportation network (e.g., with no more than 50 commodities), the size of the corresponding time-space network would rapidly grow, consisting of hundreds of vertices and arcs. The **MCIP** formulation then becomes large and extremely memory-consuming, which renders this formulation and the associated exact algorithms impractical. Therefore, we do not choose this formulation or the exact algorithms in the literature and instead use the simple path enumeration algorithm to solve small-scale instances exactly. We focus on developing an approximation algorithm based on local search heuristics to solve general instances.

4.2. The Lagrangian Dual Bound

We consider a Lagrangian relaxation problem as follows. Relax Constraints (5) and (7). For each Constraint (5), we first multiply w_s on both sides of the equation and then define λ_v^s to be the dual variable for this equivalent constraint. Define γ_e to be the dual variable for Constraint (7) for all $e \in \bar{\mathcal{A}}_{service}^C$. This modification is necessary because of the inherently different scales of dual variables. Moreover, we write $\lambda_{(o_s, t_s^o)}^s$ as λ_{source}^s and write $\lambda_{d_s}^s$ as λ_{sink}^s for simplicity.

In the incremental discount case, the cost function $c_e(x)$ of local arc $e \in \bar{\mathcal{A}}_{service}^L$ is piecewise linear concave, which as will be seen in Online Appendix A significantly simplifies solving the Lagrangian relaxation problem. In the all-unit discount case, however, the original LTL rate functions $c_e(x)$ in general are neither convex nor concave, which imposes significant challenges on the optimization of the Lagrangian relaxation problem. Thus, we use a piecewise linear concave function denoted by $\tilde{c}_e(x)$ to approximate the cost function $c_e(x)$ of each local service arc $e \in \bar{\mathcal{A}}_{service}^L$ from below. Specifically, $\tilde{c}_e(x) = \min_{1 \leq l \leq K_e} \{\tilde{r}_e^l x + \tilde{b}_e^l\}$ and $\tilde{c}_e(x) \leq c_e(x)$ for all feasible flow x of problem (CP) on arc e , where K_e is the number of segments of $\tilde{c}_e(x)$, $\tilde{r}_e^l$ and $\tilde{b}_e^l$ are the slope and intercept of segment l , respectively. For the example all-unit discount cost function in Figure 2, such an approximate function indicated by the solid lines is shown in Figure 4.

We define the Lagrangian relaxation function, where the original cost functions are replaced by the underestimate functions $\tilde{c}_e(x)$ in the all-unit discount case and $\tilde{c}_e(x) \equiv c_e(x)$ in the incremental discount case.

$$\begin{aligned} L(\mathbf{x}, \mathbf{y}, \lambda, \gamma, \mathcal{S}) = & \sum_{s \in \mathcal{S}} w_s (\lambda_{sink}^s - \lambda_{source}^s) \\ & + \sum_{e=(v_1, v_2) \in \bar{\mathcal{A}}_{service}^C} [(f_e - U_e \gamma_e) y_e + \sum_{s \in \mathcal{S}} w_s (\lambda_{v_1}^s - \lambda_{v_2}^s + \gamma_e) x_e^s] \\ & + \sum_{e=(v_1, v_2) \in \bar{\mathcal{A}}_{service}^L} [\tilde{c}_e(\sum_{s \in \mathcal{S}} w_s x_e^s) + \sum_{s \in \mathcal{S}} w_s (\lambda_{v_1}^s - \lambda_{v_2}^s) x_e^s] \\ & + \sum_{e=(v_1, v_2) \in \bar{\mathcal{A}}_{ground}} \sum_{s \in \mathcal{S}} w_s (\bar{h}_e^s + \lambda_{v_1}^s - \lambda_{v_2}^s) x_e^s \\ & + \sum_{e=(v_1, v_2) \in \bar{\mathcal{A}}_{delivery}} \sum_{s \in \mathcal{S}} w_s (\bar{u}_e^s + \lambda_{v_1}^s - \lambda_{v_2}^s) x_e^s \\ & + \sum_{e=(v_1, v_2) \in \bar{\mathcal{A}}_{dummy}} \sum_{s \in \mathcal{S}} w_s (\bar{R}_s + \lambda_{v_1}^s - \lambda_{v_2}^s) x_e^s, \end{aligned} \quad (10)$$

where $\bar{h}_e^s := h_e^s / w_s$, $\bar{u}_e^s := u_e^s / w_s$, and $\bar{R}_s := R_s / w_s$.

We define the dual function as the optimum of the Lagrangian relaxation problem:

$$\Psi(\lambda, \gamma, \mathcal{S}) = \min_{\mathbf{x}, \mathbf{y}} L(\mathbf{x}, \mathbf{y}, \lambda, \gamma, \mathcal{S})$$

$$\begin{aligned} \text{(RP)} \quad s.t. \quad & x_e^s \leq y_e, \quad \forall e \in \bar{\mathcal{A}}_{service}^C, s \in \mathcal{S}; \\ & x_e^s, y_e \in \{0, 1\}, \quad \forall e \in \bar{\mathcal{A}}, s \in \mathcal{S}. \end{aligned}$$

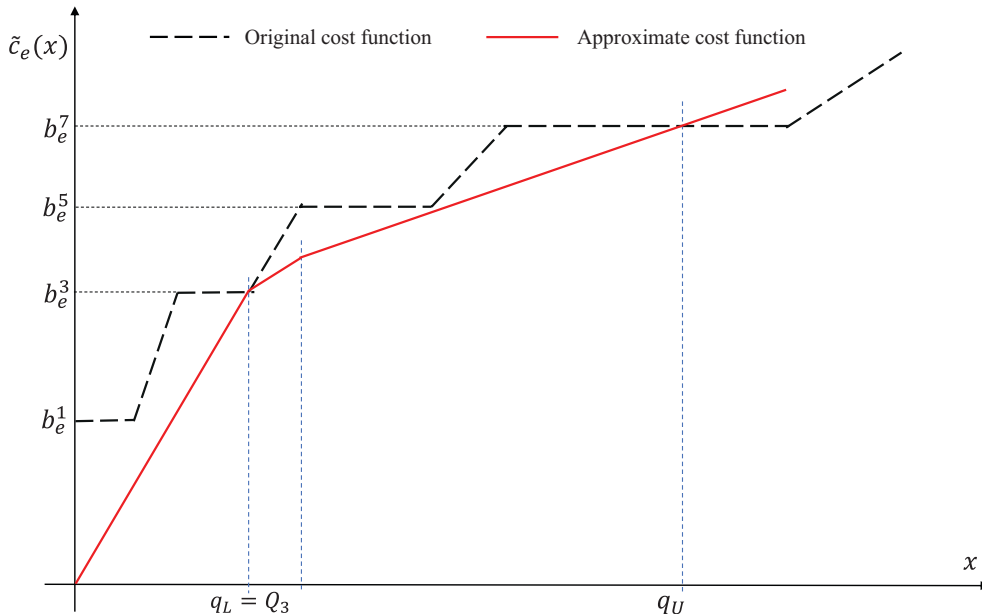
The optimization of $L(\mathbf{x}, \mathbf{y}, \lambda, \gamma, \mathcal{S})$ is decomposable on all arcs $e \in \bar{\mathcal{A}}$ and hence much easier to solve than problem (CP).

The Lagrangian dual problem is

$$\Psi(\mathcal{S}) = \max_{\lambda, \gamma} \Psi(\lambda, \gamma, \mathcal{S})$$

$$\begin{aligned} \text{(DP)} \quad s.t. \quad & U_e \gamma_e \leq f_e, \quad \forall e \in \bar{\mathcal{A}}_{service}^C; \\ & \gamma_e \geq 0, \quad \forall e \in \bar{\mathcal{A}}_{service}^C. \end{aligned} \quad (11)$$

Figure 4. (Color online) Illustration of Approximating All-Unit Discount Cost Function



Dual variables γ_e are the shadow prices of pooling service arcs' capacity. The dual Constraints (11) impose limits on the shadow prices. With $f_e - U_e \gamma_e \geq 0$, the optimal relaxation solution of y_e naturally satisfies $y_e \in \{0, 1\}$, which makes the binary requirement in problem (RP) redundant. More importantly, this is necessary when we linearize the relaxation problem to design a dual rule cost allocation in Online Appendix B.

Denote the optimal Lagrangian dual solution by (λ^*, γ^*) . Clearly, $\Psi(\mathcal{S})$ gives a lower bound of problem (CP). The relative duality gap is equal to $\Delta_{gap} := (\Gamma(\mathcal{S}) - \Psi(\mathcal{S})) / \Psi(\mathcal{S})$. Because problem (CP) is nonconvex, the duality gap is generally positive.

In the following, we introduce a method to determine approximate functions $\tilde{c}_e(x)$ in Section 4.2.1 and describe the subgradient search algorithm to solve the dual problem in Section 4.2.2.

4.2.1. The Approximation of All-Unit Discount LTL

Rate. Croxton, Gendron, and Magnanti (2003) propose a segment reduction procedure to simplify the all-unit discount LTL cost functions by generating a lower bound function with much fewer segments. However, their procedure does not guarantee that the approximate cost function is concave or minimize the approximation gap. Different from their method, we develop an optimization model to find $\tilde{c}_e(x) = \min_{1 \leq l \leq K_e} \{\tilde{r}_e^l x + \tilde{b}_e^l\}$ by minimizing the underestimation gap. We may restrict $K_e \leq K$ when searching for $\tilde{c}_e(x)$. Before introducing the optimization model, we point out the following properties of this cost function approximation problem.

First, to ensure $\tilde{c}_e(x)$ to be a lower bound, we only need the constraints $\tilde{c}_e(Q_l) \leq c_e(Q_l)$ for all points Q_l (Croxton, Gendron, and Magnanti 2003). *Second*, if we can estimate a lower bound of flows on arc e to be some $q_L = Q_{k_L}$ for any feasible solution of (CP), where k_L is a flat segment, we can ignore the approximation gap of $\tilde{c}_e(x)$ in $x < Q_{k_L}$; otherwise, we simply set $q_L = 0$ and $k_L = 1$. Then, we replace the previous k_L segments of $c_e(x)$ by a single segment with extreme points $(0, 0)$ and $(Q_{k_L}, c_e(Q_{k_L}))$. *Third*, if we can also predict an upper bound of flows on arc e to be some q_U for any feasible solution of (CP), we may further ignore the approximation gap for $x > q_U$. Find the segment $k_U = \arg\min_l \{Q_l \geq q_U : 1 \leq l \leq K\}$ and reset $Q_{k_U} = q_U$ for the approximation model. The bounds q_L and q_U can be identified by checking which freights would possibly route through the arc in any feasible solution. *Finally*, the largest approximation gap $\max_{q_L \leq x \leq q_U} \{c_e(x) - \tilde{c}_e(x)\}$ occurs at some indifference point (Croxton, Gendron, and Magnanti 2003). Thus, it is sufficient to minimize the gaps over a finite set of points. We simply select all the indifference points in the range $[q_L, q_U]$.

Let l_b and l_i be the index of flat segments (an odd integer) and the index of increasing segments (an even

integer), respectively; that is, Q_{l_b} is a break point and Q_{l_i} is an indifference point. To further simplify the model, we restrict $\tilde{c}_e(x)$ in the following way: segment l is indexed such that $\tilde{r}_e^l$ is decreasing while $\tilde{b}_e^l$ is increasing; the adjacent two segments can coincide, whereas for each break point Q_{l_b} , the underlying segments l_b and $l_b + 1$ must coincide. As the lower bound Q_{k_L} is a break point and the slope variable $\tilde{r}_e^{k_L}$ is already fixed to be $r_e^{k_L}$, the coincidence requirement for break points only applies to $l_b \geq k_L + 1$.

An example of such an approximate cost function is illustrated in Figure 4. By construction, the closest segment of $\tilde{c}_e(x)$ to each segment l of $c_e(x)$ is just l , and hence $\tilde{c}_e(Q_{l_i}) = \tilde{r}_e^{l_i} \cdot Q_{l_i} + \tilde{b}_e^{l_i}$. Then, the total estimation gaps become $\sum_{k_L \leq l_i \leq k_U} [c_e(Q_{l_i}) - \tilde{c}_e(Q_{l_i})] = \sum_{k_L \leq l_i \leq k_U} c_e(Q_{l_i}) - \sum_{k_L \leq l_i \leq k_U} (Q_{l_i} \cdot \tilde{r}_e^{l_i} + \tilde{b}_e^{l_i})$.

Because of these requirements, the general model that minimizes the approximation gap by a piecewise linear concave function can be formally defined as a linear program.

$$\max \sum_{k_L \leq l_i \leq k_U} (Q_{l_i} \cdot \tilde{r}_e^{l_i} + \tilde{b}_e^{l_i}) \quad (12)$$

$$(\text{LCAP}) \quad \text{s.t.} \quad \tilde{r}_e^l = r_e^{k_L+1}, \quad \forall 1 \leq l \leq k_L;$$

$$\tilde{b}_e^l = 0, \quad \forall 1 \leq l \leq k_L; \quad (13)$$

$$\tilde{r}_e^{k_L} \cdot Q_{k_L} + \tilde{b}_e^{k_L} = \tilde{r}_e^{k_L+1} \cdot Q_{k_L} + \tilde{b}_e^{k_L+1}; \quad (14)$$

$$\tilde{r}_e^l \geq \tilde{r}_e^{l+1}, \quad \forall 1 \leq l \leq k_U - 1; \quad (15)$$

$$\tilde{b}_e^l \leq \tilde{b}_e^{l+1}, \quad \forall 1 \leq l \leq k_U - 1; \quad (16)$$

$$\tilde{r}_e^l \cdot Q_l + \tilde{b}_e^l \leq c_e(Q_l), \quad \forall k_L + 1 \leq l \leq k_U; \quad (17)$$

$$\tilde{r}_e^{l_b} = \tilde{r}_e^{l_b+1}, \quad \forall k_L + 1 \leq l_b \leq k_U - 1, l_b \text{ is odd}; \quad (18)$$

$$\tilde{b}_e^{l_b} = \tilde{b}_e^{l_b+1}, \quad \forall k_L + 1 \leq l_b \leq k_U - 1, l_b \text{ is odd}; \quad (19)$$

$$Q_{l_i-1} \leq \frac{\tilde{b}_e^{l_i+1} - \tilde{b}_e^{l_i}}{\tilde{r}_e^{l_i} - \tilde{r}_e^{l_i+1}} \leq Q_{l_i+1}, \quad (20)$$

$$\forall k_L + 1 \leq l_i \leq k_U - 1, l_i \text{ is even};$$

$$\tilde{r}_e^l, \tilde{b}_e^l \geq 0, \quad \forall 1 \leq l \leq k_U.$$

In this model, Constraints (12) and (13) initialize the first k_L segments; Constraints (14) enforce continuity between segments k_L and $k_L + 1$; Constraints (15) and (16) order the indices of segments such that the slopes are decreasing while the intercepts are increasing; Constraints (17) enforce $\tilde{c}_e(x) \leq c_e(x)$ for all Q_l ; Constraints (18) and (19) state that for each break point Q_{l_b} the underlying segments l_b and $l_b + 1$ must coincide; and Constraints (20) are linear, because Q_{l_i-1} and Q_{l_i+1} are constants and require that the two segments l_i and $l_i + 1$ underlying the indifference point Q_{l_i} must intersect in the range $[Q_{l_i-1}, Q_{l_i+1}]$. This type of constraint prevents cases such as $\tilde{b}_e^{l_i+1} = \tilde{b}_e^{l_i}$, whereas $\tilde{r}_e^{l_i} \neq \tilde{r}_e^{l_i+1}$.

After solving for the optimal $(\tilde{r}_e^k, \tilde{b}_e^k)_{1 \leq k \leq K_e}$, redefine K_e to be the number of unique segments in $\tilde{c}_e(x)$. We claim that this approximation is indeed a lower bound and satisfies concavity.

Lemma 1. *In the all-unit discount case, the approximate cost function $\tilde{c}_e(x)$ derived from model (LCAP) satisfies piecewise linear concavity and is a lower bound of $c_e(x)$ for all $x \in [q_L, q_U]$.*

4.2.2. The Subgradient Search Algorithm. Given the dual variable (λ, γ) , the optimal solution to relaxation problem (RP) is denoted by $(\mathbf{x}(\mathcal{L}), \mathbf{y}(\mathcal{L}))$. The relaxation subproblems are all easy. The detailed solution procedure is presented in Online Appendix A.

Note that $\delta_\lambda(v, s) := w_s q_v^s + w_s \sum_{e \in \mathcal{O}(v)} x_e^s(\mathcal{L}) - w_s \sum_{e \in \mathcal{I}(v)} x_e^s(\mathcal{L})$ is a subgradient of the dual function $\Psi(\lambda, \gamma, S)$ at λ_v^s , and $\delta_\gamma(e) := \sum_{s \in S} w_s x_e^s(\mathcal{L}) - U_e y_e(\mathcal{L})$ is a subgradient at γ_e . Problem (DP) can be solved by the dual subgradient search (DSS) method. The pseudocode of DSS is described in Algorithm 3.

Algorithm 3 (Pseudocode of Algorithm DSS)

- 1: Input an initial feasible dual solution (λ, γ)
- 2: Set the best solution $(\lambda^*, \gamma^*) = (\lambda, \gamma)$, $\Psi^* = -\infty$, and $j = 1$
- 3: **while** the stopping criterion is not satisfied **do**
- 4: Solve the relaxation problem with an optimal solution $(x_e^s(\mathcal{L}), y_e(\mathcal{L}))_{s \in S, e \in \mathcal{A}}$
- 5: **if** $\Psi^* < \Psi(\lambda, \gamma, S)$ **then**
- 6: Update $(\lambda^*, \gamma^*) \leftarrow (\lambda, \gamma)$ and $\Psi^* \leftarrow \Psi(\lambda, \gamma, S)$
- 7: **end if**
- 8: Compute the subgradients $\delta_\lambda(v, s)$ and $\delta_\gamma(e)$ and the step size M_j
- 9: Update dual solution $\lambda_v^s \leftarrow \lambda_v^s + M_j \cdot \delta_\lambda(v, s) / \|\delta_\lambda\|$, $\gamma_e \leftarrow \min\{(\gamma_e + M_j \cdot \delta_\gamma(e))^+, f_e / U_e\}$, and set $j \leftarrow j + 1$
- 10: **end while**
- 11: **Return** (λ^*, γ^*)

The structure of Algorithm 3 basically follows from Wolsey (1998), except that we keep track of the best dual solution and set it as the final solution. The algorithm operates as follows.

i. **Initialization.** In line 1, a simple initial solution may be set as $(\lambda, \gamma) = \mathbf{0}$. In fact, we can find a better initial solution by applying the Poljak step size rule, which is well known for its fast convergence rate (Poljak 1987). However, the Poljak rule works well only if we know the dual optimum $\Psi(S)$ in advance or at least have a close estimation of $\Psi(S)$. For our problem, it is difficult to estimate $\Psi(S)$ because of the combinatorial nature of the primal problem (CP). Nevertheless, we may first run the subgradient search algorithm with the Poljak step size for a finite number of iterations only for initialization. Under the Poljak rule, λ and γ are

iteratively updated as

$$\begin{aligned}\lambda_v^s &\leftarrow \lambda_v^s + g_\lambda \cdot \delta_\lambda(v, s); \\ \gamma_e &\leftarrow \min\{(\gamma_e + g_\gamma \cdot \delta_\gamma(e))^+, f_e / U_e\},\end{aligned}$$

where the step sizes are

$$\begin{aligned}g_\lambda &= \left(\frac{(1 - \alpha_{est})\Gamma(S) - \Psi(\lambda, \gamma, S)}{\|\delta_\lambda\|_2^2} \right)^+; \\ g_\gamma &= \left(\frac{(1 - \alpha_{est})\Gamma(S) - \Psi(\lambda, \gamma, S)}{\|\delta_\gamma\|_2^2} \right)^+.\end{aligned}$$

The number α_{est} is considered as an estimation of the relative duality gap, which is determined through trial and error.

ii. **Subgradient search.** In lines 8 and 9, the dual variables are updated along a subgradient direction. The step size is given by $M_j = M_0 / \sqrt{j}$, for all $j \geq 1$, where M_0 is a predetermined small number. The updated value of γ_e should be nonnegative and not exceed the upper bound f_e / U_e , as explained previously.

iii. **Stopping criterion.** Although the basic subgradient search algorithm guarantees convergence, the convergence might be oscillating and too slow to be of practical value (Wolsey 1998). Thus, in lines 5–7, we track the best solution and also record the number of consecutively nonimproving iterations. The algorithm stops when $(\|\delta_\lambda\|, \|\delta_\gamma\|) = 0$ (i.e., no duality gap), or the search time reaches a predetermined limit denoted by $MAXD$, or the number of nonimproving iterations exceeds a predetermined limit denoted by NIP .

The control parameters M_0 , $MAXD$, and NIP are also empirically determined. M_0 is usually set small, whereas NIP is expected to be large. If after NIP iterations the best dual solution is still not improved, it is reasonable to believe that any further substantial improvement is unlikely.

5. The Cost Allocation Game

At the end of the planning horizon, the 3PL allocates the total shipping cost among the freights delivered through the CD-PL network. The review of Basso et al. (2019) highlights that collusion, that is, coalitional stability, is a key practical topic in collaborative logistics. This is why the core from cooperative game theory originally proposed by Gillies (1959) is an essential notion of fair allocation in the literature of this field (Agarwal and Ergun 2010, Guajardo and Rönnqvist 2016). Therefore, we choose to model the cost allocation problem as a cooperative game instead of applying mechanism design theory.

Define a cooperative game (S, Γ) , where the freights are the players. The value $\Gamma(\mathcal{R})$ of coalition $\mathcal{R}$ is defined to be the optimum of problem (CP) restricted to

freights $s \in \mathcal{R}$. An allocation μ of the system cost $\Gamma(\mathcal{S})$ to each freight is in the core $\mathcal{C}(\mathcal{S}, \Gamma)$ of the game $(\mathcal{S}, \Gamma)$, if and only if the *budget balance* constraint $\sum_{s \in \mathcal{S}} \mu_s = \Gamma(\mathcal{S})$, and the *stability* constraints $\sum_{s \in \mathcal{R}} \mu_s \leq \Gamma(\mathcal{R})$ for all $\mathcal{R} \subseteq \mathcal{S}$, are both satisfied. If the allocation μ is not budget balanced, there is an *underallocation* equal to $\Gamma(\mathcal{S}) - \sum_{s \in \mathcal{S}} \mu_s$.

A cooperative game is proper only if the coalition value function $\Gamma(\cdot)$ is subadditive, so that it is always profitable to form large coalitions (Marinakis, Migdalas, and Pardalos 2008). Clearly, because the unit LTL transportation cost decreases as volume increases and multiple TLs on a pooling arc are allowed simultaneously, subadditivity can easily be verified.

Lemma 2. *Under both incremental and all-unit discount structures, the cooperative game $(\mathcal{S}, \Gamma)$ is subadditive, that is, $\Gamma(\mathcal{R}) + \Gamma(\mathcal{U}) \geq \Gamma(\mathcal{R} \cup \mathcal{U})$, for all $\mathcal{R}, \mathcal{U} \subset \mathcal{S}$ and $\mathcal{R} \cap \mathcal{U} = \emptyset$.*

Unfortunately, as in most practical transportation problems (Özener, Ergun, and Savelsbergh 2013), a core allocation of $(\mathcal{S}, \Gamma)$ may not exist, as shown by a small counterexample.

Theorem 2. *Under both incremental and all-unit discount structures, the core of the cooperative game $(\mathcal{S}, \Gamma)$ is generally empty.*

However, the emptiness of the core does not necessarily prevent cooperation (Özener and Ergun 2008, Hezarkhani, Slikker, and Van Woensel 2016, Osicka, Guajardo, and van Oost 2020). An approximation to the core by slightly relaxing the stability constraint, namely the ϵ -core, which was proposed by Shapley and Shubik (1966), is also widely accepted as a practical fair allocation (Altan and Özener 2019). We define the ϵ -core of the shippers' game as the set of allocations μ satisfying: $\sum_{s \in \mathcal{S}} \mu_s = \Gamma(\mathcal{S})$; and $\sum_{s \in \mathcal{R}} \mu_s \leq (1 + \epsilon) \Gamma(\mathcal{R})$ for all $\mathcal{R} \subset \mathcal{S}$. An ϵ -core allocation is always budget balanced and may violate the stability relatively by at most ϵ ; that is, ϵ is the maximal relative deviation. An allocation is considered fairer than another allocation if its relative deviation ϵ is smaller.

Furthermore, we restrict our attention to the ϵ -core allocations that satisfy the following two fundamental conditions. We say allocation μ of game $(\mathcal{S}, \Gamma)$ satisfies *individual rationality* (IR) if $\mu_s \leq R_s$ for all $s \in \mathcal{S}$, which is implied in coalitional stability. If IR is violated, the shipper would immediately remove this freight from consolidation and use the direct shipping option. Moreover, allocation μ satisfies the *null-player property* (NP) if $\mu_s = R_s$, for every member s in the set $\{s \in \mathcal{S} : \Gamma(\mathcal{R} \cup \{s\}) = \Gamma(\mathcal{R}) + R_s, \forall \mathcal{R} \subseteq \mathcal{S}\}$. The NP property ensures that if a player has no contribution to the system, he should receive zero benefits, as no one would like giving away their jointly created savings to null players (Hezarkhani, Slikker, and Van Woensel 2019).

Although ϵ -core allocations may theoretically cause deviation by some subcoalition, it does not necessarily mean that this coalition would actually be formed in real life. The use of the approximate core with a small relative deviation ϵ can be justified as follows.

Remark 3. Özener and Ergun (2008) argue that, in real life, because of insufficient information sharing, limited rationality of the players, the cost of organizing collaboration, and contractual agreements, many subcoalitions cannot actually be formed. Hezarkhani, Slikker, and Van Woensel (2016) further highlight that, a positive value of ϵ reflects the fact that collaborative transportation requires investment, which includes establishing standardized information systems and perhaps using a transactional intermediary to ensure privacy of information. This is especially true in our problem. The 3PL must use a sophisticated information system to manage shipper consortium including freight routing, tracking, handling, and billing (Robinson 2016). Hence, the 3PL usually prefers to manage a large-size coalition of freights so as to be profitable.

Moreover, although the shippers collaborate in multiple planning horizons, freights rarely overlap between consecutive planning horizons, and hence the cost allocation games are generally not identical across planning horizons. $\square$

An ϵ -core allocation is determined in two steps. First, in Section 5.1, we use the Lagrangian dual to develop a stable cost allocation rule possibly with underallocation. Second, in Section 5.2, we design a budget-covering procedure to resolve the underallocation of the Lagrangian dual rule. We later computationally evaluate the stability deviation of the allocation.

5.1. The Lagrangian Dual Rule

We use the optimal Lagrangian dual solution (λ^*, γ^*) to design a cost allocation. Note that $\lambda_v^{s,*}$ is the shadow price for freight s visiting vertex v and γ_e^* is the shadow price for capacity on pooling arc $e \in \mathcal{A}_{\text{service}}$. Thus, if the duality gap is zero, $w_s(\lambda_{\text{sink}}^{s,*} - \lambda_{\text{source}}^{s,*})$ would be the payment of freight s . However, the last five terms of $L(\mathbf{x}, \mathbf{y}, \lambda^*, \gamma^*, \mathcal{S})$ in (10) are typically negative. Intuitively, all slack should also be allocated among the freights to reward their contributions to consolidation.

Because of the nonlinear TL rates and LTL rates, the slack on all service arcs $e \in \mathcal{A}_{\text{service}}$ is not readily separable by freight. Meanwhile, the slack on the ground, delivery, and dummy arcs is separable by freight. In Online Appendix B, we reformulate the nonlinear relaxation subproblems on service arcs as linear programming (LP) models and prove in Lemmas B.1 and B.2 that the linearization is exact. Then, we construct the dual problems of the linear programs. The optimal solutions of these LP dual models in fact provide an

allocation of the slack on service arcs. Define $(-\pi_e^{s,D})_{s \in \mathcal{S}}$ to be the LP dual allocation of slack for each arc $e \in \bar{\mathcal{A}}_{\text{service}}$, where $\pi_e^{s,D} \geq 0$.

The linearization in Online Appendix B is exact in the sense that the total allocation $\sum_{s \in \mathcal{S}} (-\pi_e^{s,D})$ is equal to the slack on each service arc e . The linearization can also be computed in polynomial time and does not require generating a large number of new variables as in the literature. Hence, it is efficiently computable even for very large-scale instances.

We now propose a *Lagrangian dual rule* cost allocation for the cooperative game $(\mathcal{S}, \Gamma)$ denoted by μ^D as

$$\begin{aligned} \mu_s^D = & w_s(\lambda_{\text{sink}}^{s,*} - \lambda_{\text{source}}^{s,*}) + \sum_{e \in \bar{\mathcal{A}}_{\text{service}}} (-\pi_e^{s,D}) \\ & + \sum_{e=(v_1, v_2) \in \bar{\mathcal{A}}_{\text{ground}}} w_s(\bar{h}_e^s + \lambda_{v_1}^{s,*} - \lambda_{v_2}^{s,*})^- \\ & + \sum_{e=(v_1, v_2) \in \bar{\mathcal{A}}_{\text{delivery}}} w_s(\bar{u}_e^s + \lambda_{v_1}^{s,*} - \lambda_{v_2}^{s,*})^- \\ & + \sum_{e=(v_1, v_2) \in \bar{\mathcal{A}}_{\text{dummy}}} w_s(\bar{R}_s + \lambda_{v_1}^{s,*} - \lambda_{v_2}^{s,*})^-, \quad \forall s \in \mathcal{S}. \end{aligned}$$

Define $\Psi(\mathcal{R})$ to be the Lagrangian dual optimum restricted to freights $s \in \mathcal{R}$. From Lemma B.3 in Online Appendix B and the fact that $\Psi(\lambda^*, \gamma^*, \mathcal{R}) \leq \Psi(\mathcal{R}) \leq \Gamma(\mathcal{R})$ for all $\mathcal{R} \subseteq \mathcal{S}$, we can now prove that the Lagrangian dual rule is a stable cost allocation for the game $(\mathcal{S}, \Gamma)$.

Theorem 3. *The Lagrangian dual rule μ^D is stable for the cooperative game $(\mathcal{S}, \Gamma)$. The total allocation of μ^D is equal to the Lagrangian dual optimum $\Psi(\mathcal{S})$.*

Clearly, if the duality gap is zero, μ^D is in the core. The time complexity of computing μ^D is mainly determined by the subgradient search process to solve for (λ^*, γ^*) , which is $O(G \cdot |\bar{\mathcal{A}}| \cdot T \cdot |\mathcal{S}|/\alpha^2)$, where α is a given tolerance for the dual optimum and G is the Lipschitz constant that depends on capacity U_e and freights' weight w_s .

5.2. The Budget Covering Procedure

The Lagrangian dual rule μ^D generally may not cover the total cost of transportation plan $(\mathbf{x}^*, \mathbf{y}^*)$. The relative underallocation (i.e., budget deficit) of μ^D is equal to the relative duality gap, that is, $(\Gamma(\mathcal{S}) - \sum_{s \in \mathcal{S}} \mu_s^D) / \sum_{s \in \mathcal{S}} \mu_s^D = \Delta_{\text{gap}}$. However, the duality gap typically is not small, as shown by the following computational experiments. An allocation with budget deficit is not sustainable, as the consortium would require substantial outside subsidies to operate. We thus design a budget covering procedure (BCP) to resolve the underallocation issue. The pseudocode in Algorithm 4 describes the structure of procedure BCP.

Algorithm 4 (Pseudocode of Procedure BCP)

1: Initialize $\mu^L = \mu^D$

- 2: Decompose μ_s^D onto the arcs of delivery path for each freight s
- 3: **for** each service arc $e \in \bar{\mathcal{A}}_{\text{service}}$ **do**
- 4: Distribute the underallocation on arc e among a subset $\mathcal{S}_0 \subset \mathcal{S}$ of freights
- 5: Add the distributions to μ^L
- 6: **end for**
- 7: Add the underallocations on dummy arcs to μ^L
- 8: **for** each freight $s \in \mathcal{S}$ **do**
- 9: Distribute the total underallocations on ground and delivery arcs of its delivery path among a subset $\mathcal{S}_0 \subset \mathcal{S}$ of freights
- 10: Add the distributions to μ^L
- 11: **end for**
- 12: **Return** μ^L

Procedure BCP recovers the budget arc by arc and operates in the following three phases.

i. **Initialization phase.** In line 1, the procedure starts with the Lagrangian dual rule allocation μ^D , which is stable while not budget balanced.

ii. **Decomposition phase.** In line 2, this phase proceeds in two steps. First, $w_s(\lambda_{\text{sink}}^{s,*} - \lambda_{\text{source}}^{s,*})$ is disaggregated onto the delivery path; that is, for each $e = [v_1, v_2] \in \bar{\mathcal{A}}$ with $x_e^{s,*} = 1$, the allocation is $w_s(\lambda_{v_2}^{s,*} - \lambda_{v_1}^{s,*})$.

Second, for each freight s , we determine the shared slack on each arc and then move all slack allocations to the arcs on its delivery path. Define Δ_e^s to be this shared slack; in other words,

$$\Delta_e^s = \begin{cases} -\pi_e^{s,D}, & \text{if } e \in \bar{\mathcal{A}}_{\text{service}}; \\ w_s(\bar{h}_e^s + \lambda_{v_1}^{s,*} - \lambda_{v_2}^{s,*})^-, & \text{if } e \in \bar{\mathcal{A}}_{\text{ground}}; \\ w_s(\bar{u}_e^s + \lambda_{v_1}^{s,*} - \lambda_{v_2}^{s,*})^-, & \text{if } e \in \bar{\mathcal{A}}_{\text{delivery}}; \\ w_s(\bar{R}_s + \lambda_{v_1}^{s,*} - \lambda_{v_2}^{s,*})^-, & \text{if } e \in \bar{\mathcal{A}}_{\text{dummy}}. \end{cases}$$

Define the sets of arcs $\mathcal{A}_s^1 := \{e \in \bar{\mathcal{A}} \setminus (\bar{\mathcal{A}}_{\text{ground}} \cup \bar{\mathcal{A}}_{\text{delivery}}) : x_e^{s,*} = 1\}$ and $\mathcal{A}_s^2 := \{e \in \bar{\mathcal{A}} : \Delta_e^s \neq 0, \text{ and } x_e^{s,*} = 0\}$ for each $s \in \mathcal{S}$. $\mathcal{A}_s^1$ is the set of service and dummy arcs on the delivery path, whereas $\mathcal{A}_s^2$ is the set of arcs not on the delivery path but with nonzero slack allocation.

The slack allocation on all arcs $e \in \mathcal{A}_s^2$ should be moved to the path. Then, for each $s \in \mathcal{S}$, compute $\sum_{e \in \mathcal{A}_s^2} \Delta_e^s$ and distribute this slack equally onto the arcs in $\mathcal{A}_s^1$ by updating

$$\begin{aligned} \Delta_e^s & \leftarrow \Delta_e^s + \frac{\sum_{e \in \mathcal{A}_s^2} \Delta_e^s}{|\mathcal{A}_s^1|}, \quad \forall e \in \mathcal{A}_s^1; \\ \Delta_e^s & \leftarrow 0, \quad \forall e \in \mathcal{A}_s^2. \end{aligned}$$

As a result, the temporary cost allocation on arc $e = [v_1, v_2] \in \bar{\mathcal{A}}$ for freight s is equal to $\eta_e^s := w_s(\lambda_{v_2}^{s,*} - \lambda_{v_1}^{s,*}) + \Delta_e^s$ if $x_e^{s,*} = 1$ and $\eta_e^s = 0$ otherwise. We have $\mu_s^D = \sum_{e \in \bar{\mathcal{A}}: x_e^{s,*}=1} \eta_e^s$ for all $s \in \mathcal{S}$.

iii. **Budget covering phase.** In lines 3–11, this phase recovers the budget on service, ground, delivery, and dummy arcs in sequence by searching for a subset $\mathcal{S}_0$

that maintains the IR condition of allocation μ^L . The underallocations are distributed among the set $\mathcal{S}_0$ of freights in proportion to their standalone cost R_s . The subset $\mathcal{S}_0$ varies at each step, depending on the current value of μ^L .

iii-a. For the currently selected service arc $e \in \bar{\mathcal{A}}_{service}$ in line 3, compute the underallocation

$$\Delta_e := \begin{cases} f_e - \sum_{s \in \mathcal{S}} \eta_e^s, & \text{if } e \in \bar{\mathcal{A}}_{service}^C; \\ c_e \left(\sum_{s \in \mathcal{S}} w_s x_e^{s,*} \right) - \sum_{s \in \mathcal{S}} \eta_e^s, & \text{if } e \in \bar{\mathcal{A}}_{service}^L. \end{cases}$$

Search for $\mathcal{S}_0 := \{s \in \mathcal{S} : x_e^{s,*} = 1, (R_s - \mu_s^L)/R_s \geq \tau_0\}$, where τ_0 is a small number in $(0, 1)$ determined by trial and error, such that $\mu_s^L + \Delta_e \cdot R_s / \sum_{s' \in \mathcal{S}_0} R_{s'} \leq R_s$ is true for all $s \in \mathcal{S}_0$. This subset consists of the consolidated freights that traverse arc e and currently have sufficient cost savings. If no such $\mathcal{S}_0$ exists, restart searching for $\mathcal{S}_0 := \{s \in \mathcal{S} : (R_s - \mu_s^L)/R_s \geq \tau_0\} \setminus \{s \in \mathcal{S} : x_{e'}^{s,*} = 1, e' \in \bar{\mathcal{A}}_{dummy}\}$. This redefined subset includes the consolidated freights with sufficient cost savings so far.

Then, for all $s \in \mathcal{S}_0$, update $\mu_s^L \leftarrow \mu_s^L + \Delta_e \cdot R_s / \sum_{s' \in \mathcal{S}_0} R_{s'}$. Choose the next service arc and repeat this step until all service arcs are exhausted.

iii-b. In line 7, for each freight $s \in \mathcal{S}$, compute the underallocation on the respective dummy arc e as $\Delta_s := (R_s - \eta_e^s) \cdot x_e^{s,*}$ and update $\mu_s^L \leftarrow \mu_s^L + \Delta_s$.

iii-c. For the currently selected freight s in line 8, compute the total underallocations on ground and delivery arcs of its path as

$$\Delta_s := \sum_{e \in \bar{\mathcal{A}}_{ground} : x_e^{s,*} = 1} (h_e^s - \eta_e^s) + \sum_{e \in \bar{\mathcal{A}}_{delivery} : x_e^{s,*} = 1} (u_e^s - \eta_e^s).$$

Try $\mathcal{S}_0 := \{s\}$. If $\mu_s^L + \Delta_s > R_s$, restart searching for $\mathcal{S}_0 := \{s' \in \mathcal{S} : (R_{s'} - \mu_{s'}^L)/R_{s'} \geq \tau_0\} \setminus \{s' \in \mathcal{S} : x_{e'}^{s',*} = 1, e' \in \bar{\mathcal{A}}_{dummy}\}$ such that $\mu_{s'}^L + \Delta_s \cdot R_{s'} / \sum_{s'' \in \mathcal{S}_0} R_{s''} \leq R_{s'}$ for all $s' \in \mathcal{S}_0$.

Then, for all $s' \in \mathcal{S}_0$, update $\mu_{s'}^L \leftarrow \mu_{s'}^L + \Delta_s \cdot R_{s'} / \sum_{s'' \in \mathcal{S}_0} R_{s''}$. Choose the next freight and repeat this step until all freights are exhausted.

The final cost allocation derived from procedure BCP is μ^L , namely the *Lagrangian-based rule*.

Theorem 4. The Lagrangian-based cost allocation rule μ^L for cooperative game $(\mathcal{S}, \Gamma)$ satisfies the budget balance, IR, and NP conditions.

The principle of budget covering is that each freight's allocation is increased from the dual rule μ_s^D as minimally as possible. In Online Appendix C, we present a small example using the network in Figure 1 to visualize the covering process of BCP.

Remark 4. The procedure is initialized with a stable rule μ^D , in the hope that a small increase would lead to minor stability deviation. During the decomposition phase, we evenly move slackness onto the delivery paths' service

arcs or dummy arcs that represent physical movement. Then, it is possible to distribute the underallocations mainly among the freights that traverse these arcs. During the budget covering phase, the underallocations on service, ground, and delivery arcs are distributed only among the freights that are delivered through the CD-PL network; all directly shipped freights are allocated with their stand-alone cost. The subset $\mathcal{S}_0$ always exists, because from Lemma 2, the system cost savings created by consolidated freights are nonnegative. During the search for subset $\mathcal{S}_0$, we use a single value of τ_0 throughout the procedure for simplicity and try several values of τ_0 . A smaller value of τ_0 allocates underallocations among more freights, while a larger value of τ_0 distributes more of the underallocations to the freights with high individual cost savings. $\square$

We evaluate the stability deviation of rule μ^L from procedure BCP by studying the following problem:

$$\max_{\mathcal{R} \subseteq \mathcal{S}} \left(\sum_{s \in \mathcal{R}} \mu_s^L - \Gamma(\mathcal{R}) \right). \quad (21)$$

It is computationally difficult to solve Problem (21) by enumerating an exponential number of coalitions, as each value function $\Gamma(\mathcal{R})$ is determined by a hard optimization problem. We propose a network flow model instead to solve this problem.

First, in the time-space network (Figure 3), for each freight $s \in \mathcal{S}$, we add a second arc linking the source (o_s, t_s^0) to the sink $\hat{d}_s$, which we call an *exclusion arc*; that is, there exist two arcs linking the source and the sink. The cost of this exclusion arc is equal to μ_s^L . Denote the set of exclusion arcs by $\bar{\mathcal{A}}_{excls}$, and update $\bar{\mathcal{A}} \leftarrow \bar{\mathcal{A}} \cup \bar{\mathcal{A}}_{excls}$.

Then, by defining the binary variables x_e^s and y_e as before, the exclusion of a freight from coalition $\mathcal{R}$ is represented by the flow through an arc $e \in \bar{\mathcal{A}}_{excls}$. The objective function of Problem (21) becomes

$$\begin{aligned} & \max_{\mathcal{R} \subseteq \mathcal{S}} \left(\sum_{s \in \mathcal{R}} \mu_s^L - \Gamma(\mathcal{R}) \right) \\ &= \max_{(x_e^s)_{e \in \bar{\mathcal{A}}_{excls}, s \in \mathcal{S}}} \left\{ \sum_{s \in \mathcal{S}} \sum_{e \in \bar{\mathcal{A}}_{excls}} \mu_s^L (1 - x_e^s) \right. \\ & \quad - \min_{(x_e^s, y_e)_{e \in \bar{\mathcal{A}}_{excls}, s \in \mathcal{S}}} \left[\sum_{e \in \bar{\mathcal{A}}_{service}^C} f_e y_e + \sum_{e \in \bar{\mathcal{A}}_{service}^L} c_e \left(\sum_{s \in \mathcal{S}} w_s x_e^s \right) \right. \\ & \quad \left. + \sum_{e \in \bar{\mathcal{A}}_{ground}} \sum_{s \in \mathcal{S}} h_e^s x_e^s + \sum_{e \in \bar{\mathcal{A}}_{delivery}} \sum_{s \in \mathcal{S}} u_e^s x_e^s + \sum_{e \in \bar{\mathcal{A}}_{dummy}} \sum_{s \in \mathcal{S}} R_s x_e^s \right] \Big\} \\ &= \sum_{s \in \mathcal{S}} \mu_s^L - \min_{(x_e^s, y_e)_{e \in \bar{\mathcal{A}}, s \in \mathcal{S}}} \left\{ \sum_{e \in \bar{\mathcal{A}}_{service}^C} f_e y_e \right. \\ & \quad + \sum_{e \in \bar{\mathcal{A}}_{service}^L} c_e \left(\sum_{s \in \mathcal{S}} w_s x_e^s \right) + \sum_{e \in \bar{\mathcal{A}}_{ground}} \sum_{s \in \mathcal{S}} h_e^s x_e^s \\ & \quad \left. + \sum_{e \in \bar{\mathcal{A}}_{delivery}} \sum_{s \in \mathcal{S}} u_e^s x_e^s + \sum_{e \in \bar{\mathcal{A}}_{dummy}} \sum_{s \in \mathcal{S}} R_s x_e^s + \sum_{e \in \bar{\mathcal{A}}_{excls}} \sum_{s \in \mathcal{S}} \mu_s^L x_e^s \right\}, \end{aligned}$$

subject to Constraints (5)–(8).

For each freight s , the flow balance Constraint (5) at the source is $\sum_{e \in \mathcal{O}(o_s, t_s^0)} x_s^e = 1$, which requires that exactly one dummy arc, one local service arc, or one exclusion arc can be selected between (o_s, t_s^0) and $\hat{d}_s$. Therefore, in the first equality of (22), the given exclusion variables $(x_e^s)_{e \in \bar{\mathcal{A}}_{\text{excls}}, s \in \mathcal{S}}$ implicitly determine a coalition $\mathcal{R}$, and the minimization over variables $(x_e^s, y_e)_{e \in \bar{\mathcal{A}}_{\text{excls}}, s \in \mathcal{S}}$ just gives the exact value of $\Gamma(\mathcal{R})$. The second equality of (22) simply moves the exclusion variables into the minimization problem and hence the maximization operator is not needed any more. Problem (21) is thus reduced to

$$\begin{aligned}
 \min \quad & \sum_{e \in \bar{\mathcal{A}}_{\text{service}}^C} f_e y_e + \sum_{e \in \bar{\mathcal{A}}_{\text{service}}^L} c_e \left(\sum_{s \in \mathcal{S}} w_s x_e^s \right) \\
 & + \sum_{e \in \bar{\mathcal{A}}_{\text{ground}}} \sum_{s \in \mathcal{S}} h_e^s x_e^s + \sum_{e \in \bar{\mathcal{A}}_{\text{delivery}}} \sum_{s \in \mathcal{S}} u_e^s x_e^s \\
 (\text{SVP}) \quad & + \sum_{e \in \bar{\mathcal{A}}_{\text{dummy}}} \sum_{s \in \mathcal{S}} R_s x_e^s + \sum_{e \in \bar{\mathcal{A}}_{\text{excls}}} \sum_{s \in \mathcal{S}} \mu_s^L x_e^s \\
 \text{s.t.} \quad & (5) - (8).
 \end{aligned}$$

Heuristic algorithm **MSLS** can hence be applied to solve model (SVP). During the iterative search process for solving problem (SVP), we record the relative deviation $\epsilon_{\text{dev}} := (\sum_{s \in \mathcal{R}} \mu_s^L - \Gamma(\mathcal{R})) / \Gamma(\mathcal{R})$ and the relative size $\rho_{\text{dev}} := |\mathcal{R}| / |\mathcal{S}|$ each time we find an unstable coalition $\mathcal{R}$. We select the largest ϵ_{dev} found to be the estimated maximum stability deviation. As in Özener, Ergun, and Savelsbergh (2013), ϵ_{dev} may still be a lower bound on the maximum stability deviation for the ϵ -core in general, which will be computationally evaluated.

An alternative ϵ -core allocation rule denoted by μ^G frequently used in the literature (Özener, Ergun, and Savelsbergh 2013, Altan and Özener 2019) is to scale up the dual rule by the duality gap; that is, $\mu_s^G = (1 + \Delta_{\text{gap}}) \cdot \mu_s^D$, for all $s \in \mathcal{S}$. As revealed by the following computational experiments, this rule would severely violate condition IR and hence is unacceptable in practice.

6. Computational Experiments

In this section, we computationally evaluate the optimality of the solution algorithms and the stability of allocation rules with randomly generated instances. We consider both small-scale and large-scale instances. The purpose of our computational study is to investigate the properties of our proposed algorithms and understand when the algorithms for shipper consortium problems work well. The computational study was performed on a desktop PC with an eight-core Intel i9-9900 CPU, X64-based processor running at 3.2 GHz and 64-GB RAM.

For small-scale instances, we use the path enumeration algorithm to solve for exact optimum solutions. To reduce the computational time, we first remove the

enumerations with only one freight on a pooling arc, because it is never optimal for a single freight to traverse a pooling arc alone. This solution can be clearly improved by reassigning the direct shipping path to this freight. Then, in each small-scale instance, the path enumeration algorithm is run in parallel on 16 threads of eight CPU cores.

MSLS and **DSS** are all single-trajectory algorithms, and hence both are running on one thread of one CPU core in each instance. For the two algorithms, we set the control parameters as: $\text{MAXP} = 30$ minutes, $\text{NUM} = 200$, $\text{MAXD} = 30$ minutes, and $\text{NIP} = 5,000$, if $|\mathcal{S}| < 100$; $\text{MAXP} = 180$ minutes, $\text{NUM} = 250$, $\text{MAXD} = 120$ minutes, and $\text{NIP} = 6,000$, if $100 \leq |\mathcal{S}| < 600$; $\text{MAXP} = 240$ minutes, $\text{NUM} = 300$, $\text{MAXD} = 180$ minutes, and $\text{NIP} = 7,500$, if $|\mathcal{S}| \geq 600$. Specially, the random perturbation size in algorithm **MSLS** decreases from six to four through the iterations. For algorithm **DSS**, we use $\alpha_{\text{est}} \in [5\%, 15\%]$ for the Poljak rule as the estimation of duality gap in the initialization phase and set $M_0 \in [2.5, 5]$ in the subgradient search phase. For procedure **BCP**, we try $\tau_0 \in [0.025, 0.2]$ when searching for subset $\mathcal{S}_0$, and select the one with the minimum average of increase in the freights' allocation compared with the dual rule μ^D .

Because the centralized problem (CP) that determines the coalition value $\Gamma(\cdot)$ is NP-hard, it could be too time-consuming if we solve the exact coalition values to evaluate the cooperation benefits and stability deviation, especially for large-scale networks. Therefore, as in other literature (Özener, Ergun, and Savelsbergh 2013, Kimms and Kozeletskyi 2016, van Zon, Spliet, and van den Heuvel 2021), we use approximations in the computational experiments. The computational results from the approximation algorithm provide estimates of the potential benefits that the system and individual shippers can receive from cooperation and a lower bound for the stability of the proposed cost allocation rules.

In the following, we first discuss the generation of random instances in Section 6.1. Then, we evaluate the solution algorithms and allocation rules in Section 6.2. Finally, we provide aggregated and separated analyses to investigate the influential factors on the algorithms in Sections 6.3 and 6.4, respectively, in order to identify the cases where our algorithms for shipper consortium work well.

6.1. Data Generation

We generated the data in our experiments as follows. First, we randomly generated CD-PL networks $(\mathcal{V}, \mathcal{A})$ in a Euclidean plane of size $[0, 5,000] \times [0, 5,000]$. Let $\mathcal{V}_C$, $\mathcal{V}_P$ and $\mathcal{V}_L$ denote the set of cross-docks, pools and origins/destinations, respectively. Let $(\mathcal{V}_H, \mathcal{A}_H)$ be the hub subnetwork consisting of only cross-docks and pools, where $\mathcal{V}_H := \mathcal{V}_C \cup \mathcal{V}_P$ and $\mathcal{A}_H$ is the set of arcs

restricted to the nodes in $\mathcal{V}_H$. The planning horizon is $T = 7$. We set a discrete value for the travel time a_e for each arc $e \in \mathcal{A}$ based on its Euclidian distance denoted by d_e as shown:

$$a_e = \begin{cases} 1, & \text{if } d_e \leq 500; \\ 2, & \text{if } 500 < d_e \leq 1,000; \\ 3, & \text{if } 1,000 < d_e \leq 2,000; \\ 4, & \text{otherwise.} \end{cases}$$

To represent practical situations, the pooling arcs always have longer travel time than the local arcs.

Second, a set $\mathcal{S}$ of freights were generated in the transportation network by randomly specifying an origin node, a destination node, a shipping weight, an available time, and a due time. The freights' weights were randomly generated from $[400, 6,000]$. To characterize the network, we define hub density $dense_H := |\mathcal{A}_H|/|\mathcal{V}_H|$, service density $dense_S := |\mathcal{A} \setminus \mathcal{A}_H|/|\mathcal{V}_H|$, and freight density $dense_F := |\mathcal{S}|/|\mathcal{A}_H|$. The service density is the average number of terminal nodes linked with the hub nodes and measures the connectivity of freights to the hubs. The freight density measures how many freights would be potentially consolidated on a pooling arc. The number of feasible delivery paths of freight s is denoted by np_s .

For ease of analysis, we only consider the instances where the maximal potential shipment flows on each pooling arc e is no more than $U_e + \delta_{\text{compete}}$, where $\delta_{\text{compete}} \ll U_e$; that is, the consolidated flow is sufficient for at most one TL, and the residual flow is limited by δ_{compete} . It is not economical to ship this amount of residual flow by TL. As a result, because of the capacity limit, not all the freights connected to this pooling arc can be selected into consolidation and they compete for the TL rate. The higher value of δ_{compete} , the more freights are potentially in competition. Thus, δ_{compete} may be interpreted as a measure of capacity competition level.

The transportation cost functions are defined as follows. TL rates on pooling arcs are randomly generated in proportion (e.g., 6 ~ 10) to the Euclidean distance of the arcs. The truck capacity is set as $U_e = 40,000$ lbs. First, we use the practical LTL rate structure to generate LTL cost functions in the experiments, which is given in dollars per hundred pounds (a.k.a. CWT or hundred weight). For the all-unit discount structure, the weight breaks are $[0, 5C]$, $[5C, 10C]$, $[10C, 20C]$, $[20C, 50C]$, $[50C, 100C]$, and $[100C, 200C]$, where C is the abbreviation for CWT or hundred weight. The unit rate r_e^l of the first increasing segment is randomly generated in proportion (e.g., 0.001 ~ 0.002) to the Euclidean distance of local arc e and the discounts r_e^{l+2}/r_e^l are uniformly generated from $[0.5, 0.9]$ and decreasing in l . Flat segments and indifference points are automatically added to ensure the continuity of the cost

function. Second, we solve model (LCAP) to generate piecewise linear concave cost functions and use these functions for the incremental discount structure instead of separately generating these functions. We record the number of segments K_e and the average of discount levels r_e^l/r_e^{l+1} denoted by ρ_{inc} for the incremental discount costs in each instance. We also compute the average approximation gap $\Delta_{\text{appr}} := (1/|\bar{\mathcal{A}}_{\text{service}}^L|) \cdot \max_{x \in [q_L, q_U]} (c_e(x) - \tilde{c}_e(x))/\tilde{c}_e(x)$ for the all-unit discount cost functions. The two cost structures are studied in the same set of instances for the purpose of comparison.

The scale of instances is defined as follows: small scale, with hub size $|\mathcal{V}_H| \in [5, 10]$ and freight size $|\mathcal{S}| \in [20, 80]$; and large scale, with $|\mathcal{V}_H| \in [20, 55]$ and $|\mathcal{S}| \in [100, 1,000]$. As shown in Table 2, in each class, we randomly generate instances by considering five factors: (1) hub density $dense_H$; (2) service density $dense_S$; (3) freight density $dense_F$; (4) delay cost, that is, lateness cost u_s and ground holding cost $h_s = u_s/2$; and (5) competition level δ_{compete} . As shown in Tables 3 and 4, we consider a low level and a high level for each factor. There are 32 combinations of these factor levels, and 10 instances for each combination, for a total of 320 instances in both small-scale and large-scale cases.

We give a summary view on the data of all randomly generated small- and large-scale instances in the experiments in Table 2. These instances cover a variety of CD-PL networks with typical network sizes, densities, freights' weight, and practical LTL rates. A visual illustration of some of these networks in the small-scale case is presented in Online Appendix D.

6.2. Algorithmic Results Analyses

To evaluate the optimality and time complexity of heuristic **MSLS**, we compute the system cost improvement compared with the noncollaboration case $\Delta_{\text{impr}} := (\sum_{s \in \mathcal{S}} R_s - \Gamma(\mathcal{S}))/\sum_{s \in \mathcal{S}} R_s$, the optimality gap compared with the exact solution $\Delta_{\text{opt}} := (\tilde{\Gamma}(\mathcal{S}) - \Gamma(\mathcal{S}))/\Gamma(\mathcal{S})$ if applicable, the Lagrangian duality gap Δ_{gap} , the average capacity utilization of trucks carrying TL shipments on pooling arcs $\rho_{\text{TL}} := \sum_{e \in \bar{\mathcal{A}}_{\text{service}}^c} \sum_{s \in \mathcal{S}} w_s x_e^{s*} / \sum_{e \in \bar{\mathcal{A}}_{\text{service}}^c} U_e y_e^*$, and the consolidation ratio $\rho_{\text{cld}} := 1 - \sum_{s \in \mathcal{S}} \sum_{e \in \bar{\mathcal{A}}_{\text{dummy}}} x_e^{s*} / |\mathcal{S}|$. We also record the running time in minutes denoted by t_{exact} and t_{MSLS} for the exact algorithm and heuristic **MSLS**, respectively. In essence, ρ_{TL} measures the degree of consolidation through the network; that is, freights are consolidated on fewer number of pooling arcs and would generate more cost savings. A high value of ρ_{cld} implies that most of the shippers' freights can be economically included in the CD-PL network.

Table 2. Summary View of the Random Instances

	Small scale				Large scale			
	Mean	Standard deviation	Minimum	Maximum	Mean	Standard deviation	Minimum	Maximum
$ \mathcal{V}_C $	4.46	0.91	3.00	6.00	16.08	4.28	10.00	30.00
$ \mathcal{V}_P $	2.99	0.72	2.00	4.00	13.11	2.41	10.00	24.00
$ \mathcal{V}_L $	26.95	8.26	14.00	46.00	155.84	59.37	80.00	374.00
$ \mathcal{A} $	42.16	11.73	19.00	79.00	278.57	92.53	144.00	640.00
$ \mathcal{S} $	41.50	17.76	20.00	80.00	384.59	167.17	160.00	1,000.00
$dense_H$	1.19	0.27	0.57	2.00	2.94	0.58	1.88	5.62
$dense_S$	4.46	1.22	2.29	6.83	6.50	1.57	4.45	8.70
$dense_F$	4.88	2.17	1.54	10.00	1.85	0.62	0.88	3.42
w_s	3,136.11	1,463.58	1,000.00	6,000.00	2,543.32	1,551.78	400.00	6,000.00
np_s	3.68	1.12	2.00	6.00	2.95	0.92	2.00	5.00
K_e	3.82	0.86	2.00	5.00	3.78	0.77	2.00	5.00
ρ_{inc}	0.44	0.31	0.00	0.85	0.42	0.22	0.00	0.91
Δ_{appr}	0.23	0.02	0.16	0.27	0.27	0.02	0.23	0.31

To evaluate the Lagrangian-based rule μ^L , we record the maximum relative deviation ϵ_{dev} ever found by heuristic **MSLS** and the relative size ρ_{dev} of the respective unstable coalition. Furthermore, for small-scale instances, we check the value of ϵ_{dev} for all coalitions $\mathcal{R}$ with $|\mathcal{R}| \geq \lfloor 0.8 \cdot |\mathcal{S}| \rfloor$, as unstable coalitions found in the experiments are mostly large. We also compute the average of the individual cost savings $\Delta_{bnf} := (1/|\mathcal{S}_{clt}|) \cdot \sum_{s \in \mathcal{S}_{clt}} (R_s - \mu_s^L)/R_s$, where $\mathcal{S}_{clt} \subseteq \mathcal{S}$ is the set of consolidated freights. Moreover, for the purpose of comparison, we evaluate the scaling-up rule μ^G by computing the relative deviation from stand-alone costs $\epsilon_{IR} := \max_{s \in \mathcal{S}} (\mu_s^G - R_s)/R_s$. All these proportional numbers Δ_{impr} , Δ_{opt} , Δ_{gap} , ρ_{TL} , ρ_{clt} , ϵ_{dev} , ρ_{dev} , Δ_{bnf} , and ϵ_{IR} are reported in percentage values.

For small-scale instances, we show the computational results of the enumeration algorithm and the performance of allocations based on the exact optimum in Figure 5, whereas the approximation results for heuristic **MSLS** are shown in Figure 6. Under both cost structures, the exact optimum derived after substantial computational time reduces system cost by more than 30% on average. We observe $\Delta_{gap} = 0$ in 31.68% and 0% of the instances for the incremental and all-unit discount cases, respectively. Thus, in these instances of the incremental discount case, the core is nonempty and rule μ^D is in the core. Furthermore, rule μ^L is stable in 83.23% and 47.52% of the instances under the incremental and all-unit discount structures, respectively, which also implies the non-emptiness of the core. The density plots of ρ_{dev} imply that the sizes of unstable coalitions are mostly 1 ~ 5 less than the number of total freights.

Moreover, as shown in Figure 6, heuristic **MSLS** is highly efficient for these small-scale instances and solves more than 80% of the instances to optimality with computational time less than two minutes. The averages of the optimality gaps under the incremental and all-unit discount structures are 0.3036% and 0.3052%,

respectively. The results show that our solution algorithm **MSLS** can find promising approximate optimum solutions for typical CD-PL networks as in our experiments. Consequently, the results of allocations based on approximate optimum solutions are close to the exact ones.

Figure 7 shows the results for large-scale instances. Because of the significant increase in network size, the computational time of searching for good solutions grows rapidly. With more freights in the network, the system cost is reduced by more than 40% in 96.18% and 94.59% of the instances for the incremental and all-unit discount structures, respectively. The duality gaps are rarely zero. Under the incremental discount structure, we have relative deviations $\epsilon_{dev} \leq 5\%$ and $\epsilon_{dev} \leq 10\%$ in 71.02% and 96.50% of the instances, respectively. Under the all-unit discount structure, we have $\epsilon_{dev} \leq 5\%$ and $\epsilon_{dev} \leq 10\%$ in 68.15% and 95.86% of the instances, respectively. The results for ρ_{dev} also imply that the sizes of most unstable coalitions are slightly smaller than the grand coalition, especially under the incremental discount structure.

These results reveal that rule μ^L typically has a much lower relative deviation and the system cost savings Δ_{impr} are higher under the incremental discount structure than the all-unit case. Note that μ^L guarantees individual rationality. The individual cost savings are significant on average, mostly more than 30% in the small-scale case and over 50% in the large-scale case. In contrast, Figure 8 shows that the scaling-up rule μ^G severely violates individual rationality in most instances by more than 5% for the small-scale case and more than 10% for the large-scale case. Thus, μ^G is not acceptable.

6.3. Aggregated Effect Analyses

We aggregatively investigate the factors' effects on the solution qualities and allocations to understand the properties of our algorithms. In Tables 3 and 4, we fix each factor level and compute the average values and

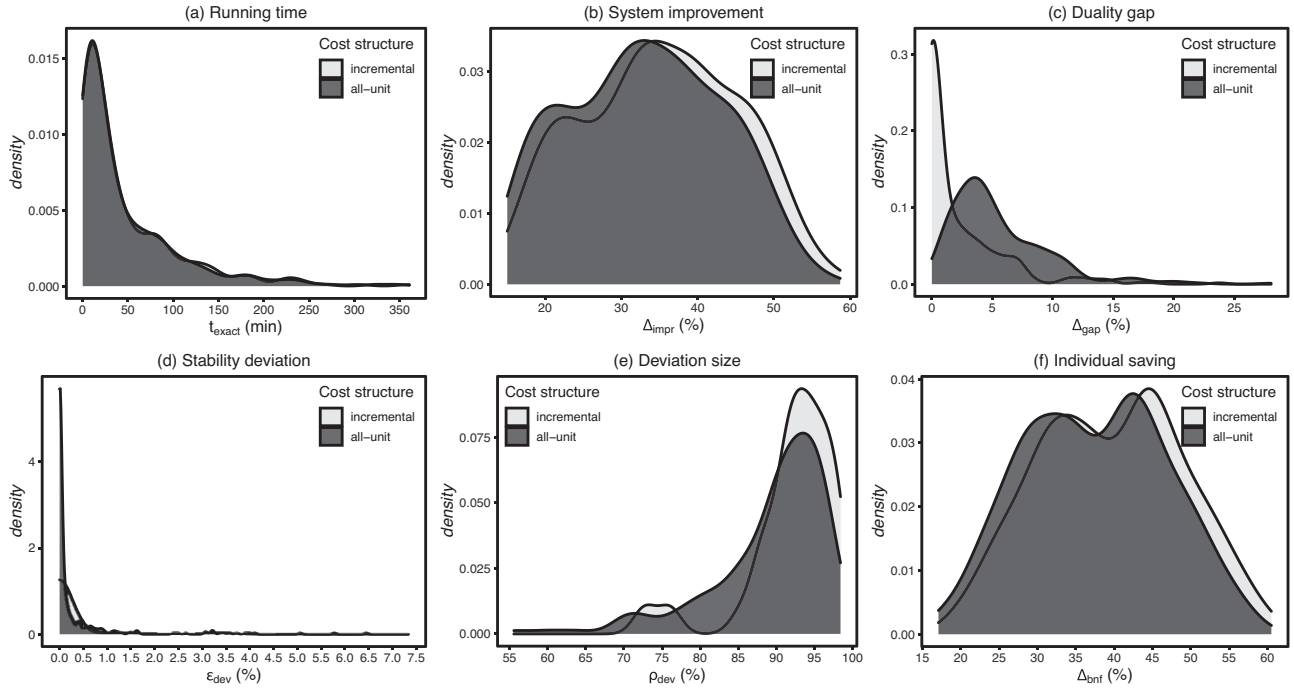
Table 3. Results for the Small-Scale Case (Unit: %)

Factor	Range	Incremental discount										All-unit discount														
		Δ_{impr}					Δ_{nop}					ϵ_{dec}					Δ_{nop}					ϵ_{dec}				
		Standard	Standard deviation	Average	Standard deviation	Average	Standard	Standard deviation	Average	Standard deviation	Average	Standard	Standard deviation	Average	Standard deviation	Average	Standard	Standard deviation	Average	Standard deviation	Average	Standard	Standard deviation	Average	Standard deviation	
$dense_H$	[0.5, 1.5]	35.54	10.58	83.54	13.83	60.83	11.23	1.76	3.03	0.32	1.06	92.73	3.18	33.56	10.42	82.81	14.19	60.88	11.28	5.02	3.57	0.38	1.01	89.28	7.02	
	(1.5, 2.5]	35.75	8.94	83.93	10.95	58.31	8.43	3.77	4.04	0.18	0.63	93.22	3.96	33.54	8.76	82.97	11.56	58.56	8.50	7.44	4.51	0.33	0.69	92.05	4.66	
$dense_S$	[2.5, 5]	28.32	8.17	76.51	15.54	51.04	9.23	0.96	3.29	0.08	0.56	80.17	8.74	26.58	7.95	75.73	15.79	51.09	9.35	3.49	3.91	0.17	0.50	81.09	8.87	
	(5, 7.0]	40.41	8.01	88.68	6.51	65.76	7.71	3.53	3.50	0.50	1.23	93.69	2.99	38.13	8.18	88.15	6.94	65.82	7.84	7.51	3.63	0.46	1.05	92.42	4.27	
$dense_F$	[1.0, 5.0]	28.52	8.48	76.96	15.26	51.06	9.30	1.17	3.40	0.11	0.59	81.62	8.05	26.77	8.26	76.18	15.54	51.12	9.42	3.74	4.02	0.23	0.66	81.81	8.88	
	(5.0, 10.0]	41.01	8.04	89.87	5.93	65.92	7.66	2.91	3.32	0.52	1.26	93.68	3.16	38.74	8.23	89.32	6.38	65.98	7.81	6.89	3.54	0.52	1.14	91.91	4.74	
$\delta_{complete}$	[0, 2500]	33.37	10.37	81.35	14.27	56.36	11.36	1.62	3.42	0.22	0.79	90.02	7.88	31.47	10.11	80.72	14.50	56.50	11.51	4.67	4.09	0.31	0.83	87.81	8.73	
	[5000, 7500]	38.10	9.05	87.12	8.97	64.02	8.52	3.14	3.41	0.44	1.20	93.45	3.33	35.86	9.22	86.46	9.57	63.90	8.60	6.91	3.68	0.43	1.02	91.15	5.10	
u_S	[0.05, 0.1]	38.05	10.49	86.45	12.39	60.95	11.56	2.62	3.81	0.40	1.11	91.91	6.78	35.98	10.36	85.92	12.57	61.02	11.72	6.18	4.45	0.43	1.03	89.06	8.16	
	(0.1, 0.2]	31.92	8.74	80.24	12.62	57.27	10.11	1.71	3.04	0.19	0.78	91.95	4.78	29.95	8.58	79.48	13.05	57.29	10.08	4.75	3.50	0.27	0.74	89.27	7.06	

Table 4. Results for the Large-Scale Case (Unit: %)

Factor	Range	Incremental discount										All-unit discount													
		Δ_{impr}					Δ_{nop}					Δ_{impr}					Δ_{nop}								
		ρ_{TL}	ρ_{ch}	Standard deviation	Average deviation	Standard deviation	ρ_{TL}	ρ_{ch}	Standard deviation	Average deviation	Standard deviation	ϵ_{dev}	ρ_{dev}	Standard deviation	Average deviation	Standard deviation	ϵ_{dev}	ρ_{dev}							
$dense_H$	[1.5, 3.0]	52.54	6.48	80.16	8.11	66.86	7.99	9.04	4.53	3.32	3.20	98.38	1.04	50.77	6.69	78.72	8.31	67.59	7.79	12.55	4.39	4.02	3.00	96.48	1.53
$dense_S$	(3.0, 6.0]	53.19	5.83	76.12	7.51	68.82	6.99	9.60	4.99	3.75	3.17	98.49	0.87	51.80	6.08	74.84	8.06	69.58	6.47	12.59	4.73	4.22	2.74	96.66	1.37
	[4.0, 8.0]	48.34	5.17	73.19	7.58	63.65	5.70	7.44	3.60	2.64	2.70	98.89	0.52	46.57	5.33	71.89	7.21	64.20	5.46	10.53	3.50	3.48	2.51	97.02	1.38
$dense_F$	(8.0, 10.0]	57.35	4.37	83.60	5.13	73.35	5.32	12.29	4.39	5.10	3.21	97.99	1.05	56.00	4.47	82.91	5.10	73.79	5.15	15.63	4.48	5.15	2.91	96.27	1.35
	[0.5, 2.0]	48.64	5.54	73.08	7.60	64.09	6.41	7.98	4.24	3.00	3.07	98.80	0.67	46.95	5.76	71.87	7.27	64.64	6.22	10.92	4.06	3.73	2.75	96.97	1.42
$\delta_{complete}$	(2.0, 4.0]	57.10	4.61	84.14	4.77	72.77	5.28	11.28	3.88	4.55	2.86	98.09	1.04	55.61	4.76	83.26	5.03	73.19	5.15	14.89	4.14	4.82	2.76	96.19	1.41
	[0, 2500]	51.84	6.54	77.99	7.89	67.20	7.29	8.66	4.14	3.19	2.72	98.72	0.74	50.14	6.75	76.68	8.33	67.80	6.95	11.92	4.11	3.90	2.64	96.74	1.37
u_s	[5000, 7500]	52.23	6.64	76.79	9.04	68.05	7.24	10.16	4.82	4.12	3.53	98.21	1.03	50.73	6.87	76.07	8.58	68.42	7.24	13.26	5.03	4.44	2.96	96.60	1.53
	[0.05, 0.1]	52.74	6.91	78.32	8.33	68.80	7.07	8.88	4.34	3.55	3.22	98.37	0.92	51.04	7.35	77.29	8.29	69.39	6.89	12.23	4.61	4.21	2.86	96.37	1.50
	(0.1, 0.2]	51.10	6.03	76.37	8.46	66.04	7.24	9.89	4.67	3.68	3.04	98.64	0.88	49.61	5.98	75.30	8.52	66.41	6.97	12.88	4.55	4.06	2.71	97.06	1.27

Figure 5. Density Curves for the Small-Scale Case



standard deviations of the performance indicators over the instances at the given factor level in the small- and large-scale cases, respectively. We use the exact optimum in the small-scale case and the approximate optimum in the large-scale case. The effects by the two algorithms are found to be almost the same, as discussed here.

As expected, under both discount structures, shipper consortium collaboration produces significantly larger system cost savings Δ_{impr} when the network sizes ($|V_H|$, $|S|$) and densities (dens_H , dens_S , dens_F) are higher, and the delay costs (u_s , h_s) are lower. The consolidation ratio ρ_{clt} and TL capacity utilization ρ_{TL} also increase when the densities are higher, except for hub density dens_H . With fewer pooling arcs in the network, the options for delivery paths are more limited, and then the TL flow volumes can be higher, which implies consolidation is economical for more freights. In contrast, higher value of hub density implies TL flow volumes are lower and less likely to be economical. This is true whenever the exact algorithm or our approximation algorithm is applied to shipper consortium problems.

The stability deviation ϵ_{dev} obviously increases when network sizes, densities dens_S and dens_F and competition level δ_{compete} are higher, or the delay costs (u_s , h_s) are lower. In these cases, TL flows are larger or more freights compete for capacity, and the duality gaps Δ_{gap} are typically higher. This makes it more difficult to recover the budget with a small stability deviation. When hub density dens_H is increased, ϵ_{dev} is lower in the small-scale case but higher in the large-

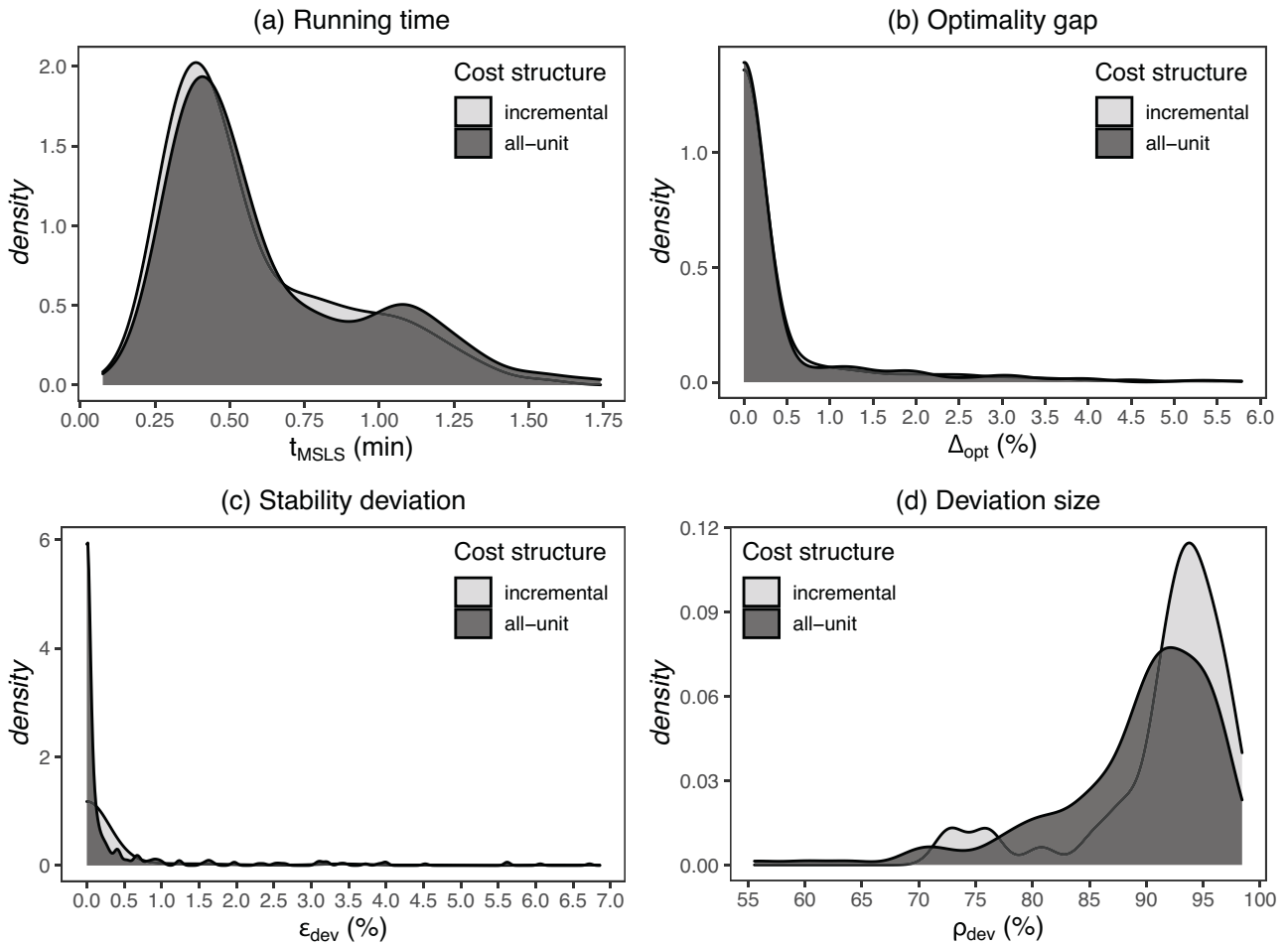
scale case. This difference is most likely due to network scale effects, as the approximation algorithm is found to be typically efficient.

6.4. Separated Effect Analyses

In this section, we separately investigate the factors' effects on the randomly generated instances to understand further when our algorithms for shipper consortium problems work well. Figures 9–11 are scatter plots showing the characteristics of interest for the small- and large-scale instances, whereas data points are differentiated between the incremental and all-unit discount cases.

In Figure 9, we show the relationships of approximation gap Δ_{appr} , capacity utilization ρ_{TL} , and duality gap Δ_{gap} with the stability deviation ϵ_{dev} , where smooth regression curves are plotted to demonstrate the trends of relationships. Clearly, the duality gap Δ_{gap} under the all-unit discount structure is typically increasing in the approximation gap Δ_{appr} , as expected. Meanwhile, larger duality gaps and capacity utilization are both associated with higher stability deviations of the allocation rule μ^L in the large-scale case. Thus, the approximation gap may also increase the stability deviation of μ^L .

In contrast, for the small-scale case, these effects on stability are only minor, as the deviations are mostly close to zero. Recall that the stability results shown in Figures 5 and 6 for the exact and approximation methods are almost the same. Thus, we believe this difference is most likely because of much more freight

Figure 6. Density Curves for Approximation Results in the Small-Scale Case

being included in the large-scale networks and hence much larger flows on the arcs than in the small-scale networks. When flows are large, the cost functions $c_e(x)$ of local arcs for such range of flows seem “more concave,” and the capacity limits on pooling arcs are often tight. As a result, the centralized optimization problem becomes “less linear.” It is well known that designing core allocations for linear network flow games is easy (Agarwal and Ergun 2008), which may explain the difference.

In Figures 10 and 11, we check under what cases stability deviation of μ^L is zero or small under the all-unit discount structure. The patterns for the incremental discount structure are quite similar and hence are not shown in this paper. In the small-scale case, we observe that μ^L is in the core for a wide variety of CD-PL networks, as shown by the wide range of network size $|\mathcal{V}|$ and densities ($dense_H$, $dense_S$, and $dense_F$). The duality gap may cause μ^L to be unstable, which is more likely when the service density is higher.

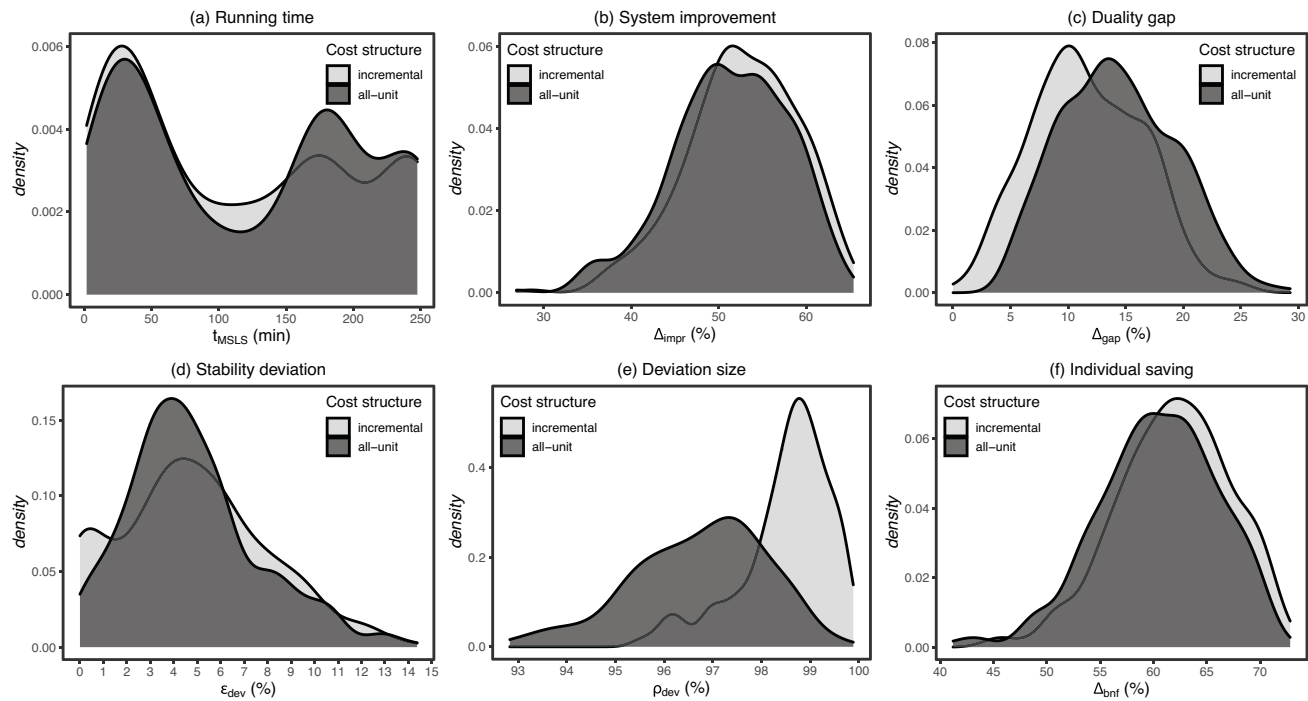
In the large-scale case, the stability deviation of rule μ^L is most likely small when the network size $|\mathcal{V}| \leq 200$,

the hub density $dense_H \leq 3$, the service density $dense_S \leq 6$, or the freight density $dense_F \leq 2$. For such networks, the capacity on pooling arcs is rarely tight, and the utilization level ρ_{TL} is moderate. Also, the duality gaps and stability deviation are typically small.

7. Concluding Remarks

This paper studies collaborative logistics via a shipper consortium operated by a 3PL. We consider both incremental discount and all-unit discount LTL rates and a flat TL cost rate in the centralized optimization problem that minimizes the total shipping cost. The problem is highly intractable, both theoretically and empirically. We develop a time-space network flow model of the problem and propose a computationally efficient approximation algorithm based on local search heuristics. Extensive computational experiments show that the shipper consortium collaboration substantially reduces the total shipping cost by more than 40% in most instances. To allocate the shipping cost fairly, we develop Lagrangian dual allocation

Figure 7. Density Curves for the Large-Scale Case



rules. In the case of all-unit discount LTL rates, we first develop a linear program to approximate the LTL rates from below by piecewise linear concave

functions with minimized estimation gap; meanwhile, the incremental discount LTL rates are already concave functions. We decompose and linearize the

Figure 8. Density Curves for the Scaling-Up Rule

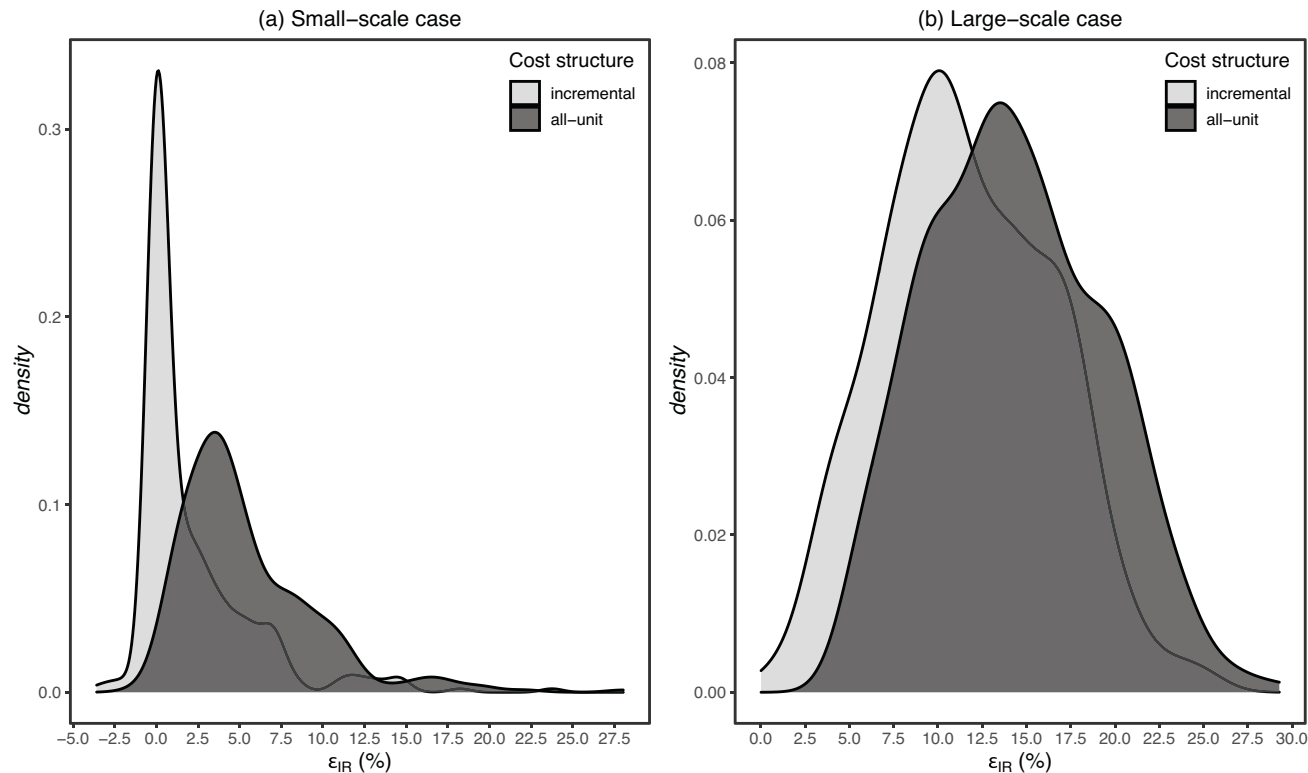
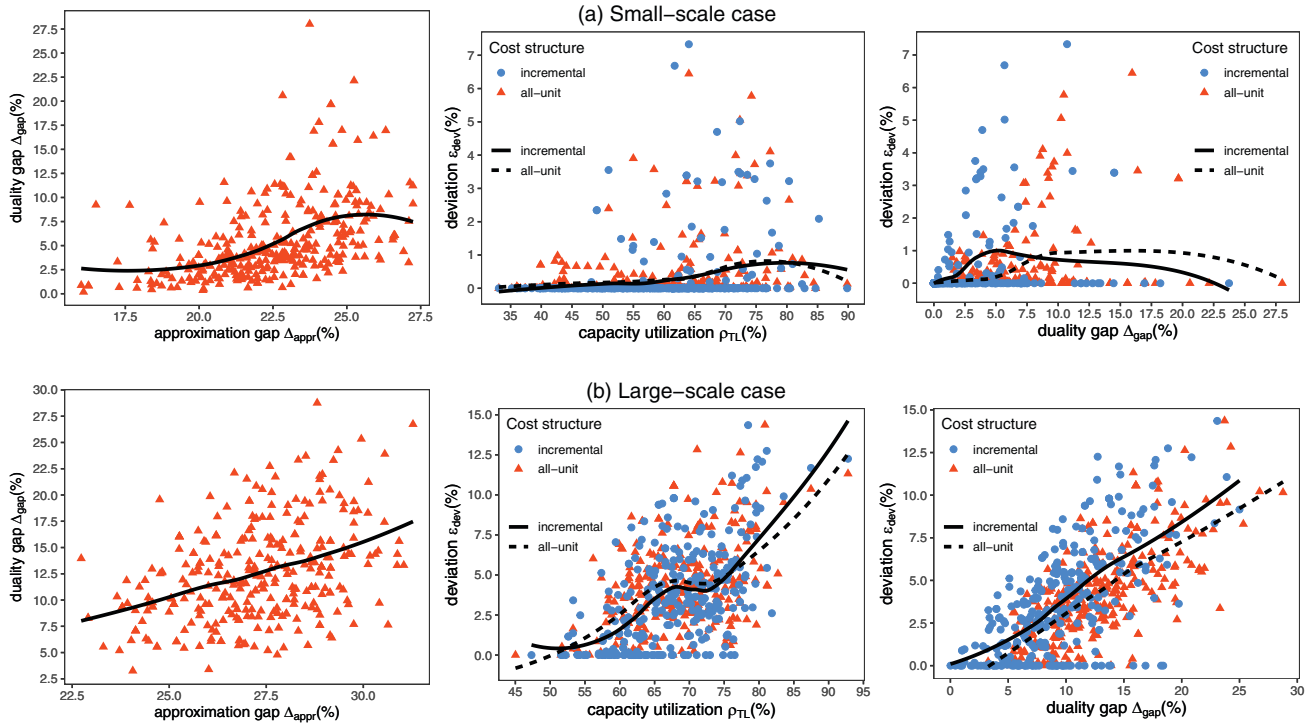


Figure 9. (Color online) Effects of Duality Gap and Stability Deviation

Lagrangian dual problem by utilizing total unimodularity and concavity. Then, we develop a stable cost allocation rule from the linearized dual models. This

Lagrangian dual rule generally has an underallocation equal to the duality gap, which may be more than 10% in relative. Finally, to cover the underallocation

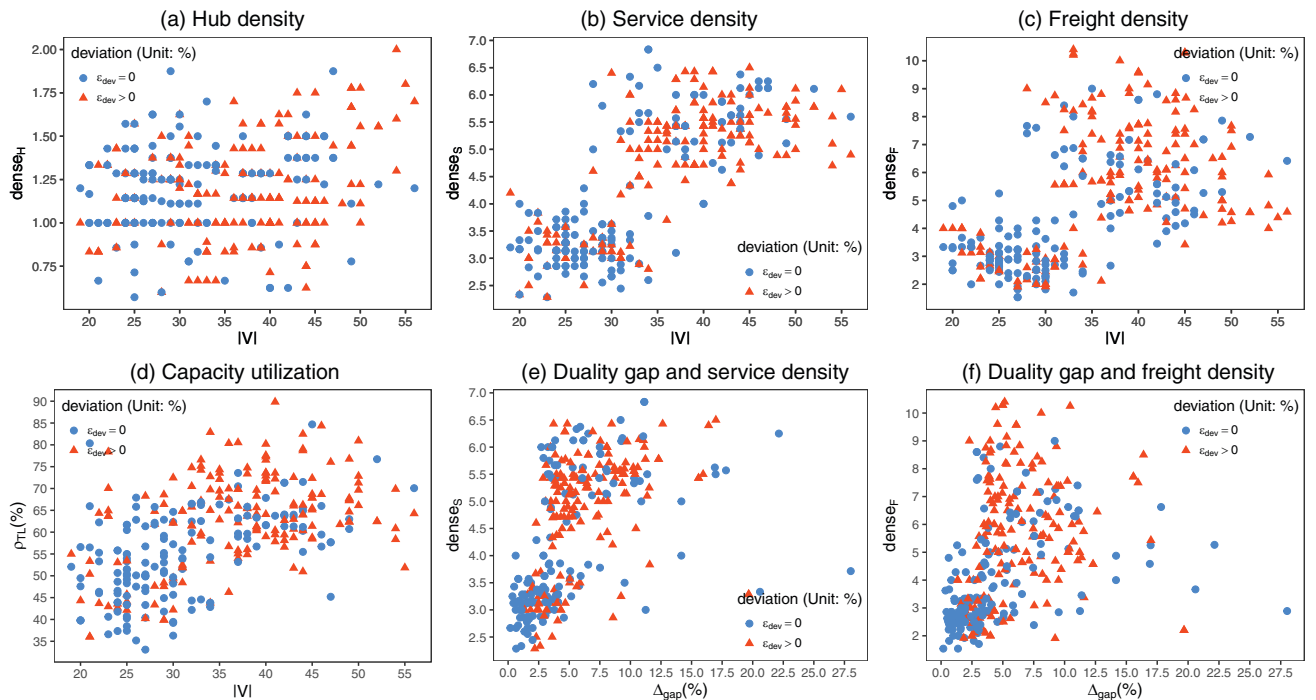
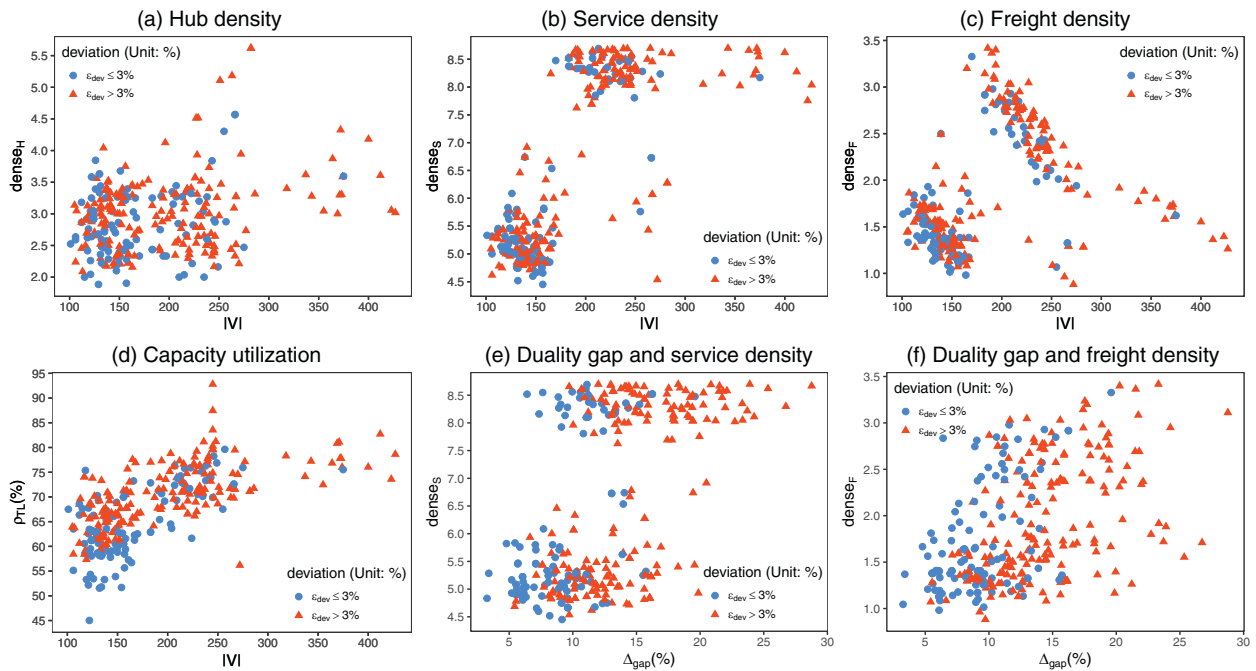
Figure 10. (Color online) Stability Analyses for the Small-Scale All-Unit Discount Case

Figure 11. (Color online) Stability Analyses for the Large-Scale All-Unit Discount Case



of the stable rule, we further develop a budget covering procedure and define an ϵ -core allocation, which satisfies individual rationality and null-player property, and attempts to violate coalitional stability as minimally as possible. Through extensive computational experiments, we find that this Lagrangian based rule is mostly in the core for small-scale networks while its relative deviation is typically no more than 5% for large-scale networks. Under this rule, consolidated freights receive individual cost savings that are more than 50% on average.

Our work provides several insights about the properties of our proposed algorithms for managing shipper consortia. First, when network sizes, service and freight densities, and capacity competition level are high, or delay costs are low, consolidation is economical for a major proportion of freights and the system cost savings derived by our optimization algorithm is typically large. However, in such cases, the stability deviation of the Lagrangian-based allocation is increased. Second, when the Lagrangian-based allocation is applied to practical CD-PL networks, the stability deviation is most likely small for networks with no more than 200 nodes, hub density no more than three, service density no more than six, or freight density no more than two. Third, our work reveals that the Lagrangian-based allocation is obviously stabler under the incremental discount structure than under the all-unit discount structure. Because of the

concavity of incremental discount structure, the system cost savings and consolidation ratio are also typically higher. Finally, when the hub density of the CD-PL network is low, system cost savings and consolidation ratio are typically high, whereas the stability deviations of the Lagrangian-based allocation are small only if the network scale is large.

Our work suggests some interesting and valuable opportunities for further research. First, it would be useful to extend our cooperative game model with shipment splitting. Because of capacity constraints, shipment splitting makes the centralized optimization problem much more difficult to solve, and hence new algorithms must be designed. Second, it would be valuable to identify other cost allocation rules that are also efficiently computable and highly stable. Our computational results reveal that the size of the most deviating coalition is often slightly smaller than the grand coalition. Hence, it might be useful to consider a relaxation of the core in which the rationality constraints of these coalitions are considered. In conclusion, we hope that our work will encourage further research on these practically important and theoretically challenging research issues.

Acknowledgments

The authors appreciate valuable comments by the editor-in-chief, special issue editor, and the two anonymous referees that helped improve the paper.

References

- Agarwal R, Ergun Ö (2008) Mechanism design for a multicommodity flow game in service network alliances. *Oper. Res. Lett.* 36(5): 520–524.
- Agarwal R, Ergun Ö (2010) Network design and allocation mechanisms for carrier alliances in liner shipping. *Oper. Res.* 58(6): 1726–1742.
- Altan B, Özener OÖ (2019) Cost allocation mechanisms in a peer-to-peer network. *Networks* 73(1):104–118.
- Arslan O, Archetti C, Jabali O, Laporte G, Speranza MG (2020) Minimum cost network design in strategic alliances. *Omega* 96: 102079.
- Basso F, D'Amours S, Rönnqvist M, Weintraub A (2019) A survey on obstacles and difficulties of practical implementation of horizontal collaboration in logistics. *Internat. Transportation Oper. Res.* 26(3):775–793.
- Caleres (2007) How to configure cross-docks and pools. Accessed March 10, 2019, https://tms.caleres.com/html/help/webhelp/en/configuration/creat_cross_docks.htm.
- CaseStack (2015) Collaborative freight consolidation programs. Accessed March 10, 2019, <http://www.casestackconsolidation.com>.
- Chabot T, Bouchard F, Legault-Michaud A, Renaud J, Coelho LC (2018) Service level, cost and environmental optimization of collaborative transportation. *Transportation Res. Part E: Logist. Transportation Rev.* 110:1–14.
- Cleophas C, Cottrill C, Ehmke JF, Tierney K (2019) Collaborative urban transportation: Recent advances in theory and practice. *Eur. J. Oper. Res.* 273(3):801–816.
- Creemers S, Woumans G, Boute R, Beliën J (2017) Tri-vizor uses an efficient algorithm to identify collaborative shipping opportunities. *Interfaces* 47(3):244–259.
- Croxton KL, Gendron B, Magnanti TL (2003) Models and methods for merge-in-transit operations. *Transportation Sci.* 37(1):1–22.
- CT Logistics (2017) CT TranSaver program—shipping co-op. Accessed March 10, 2019, <http://www.ctlogistics.com/services/shipping-co-op>.
- Diaby M, Martel A (1993) Dynamic lot sizing for multi-echelon distribution systems with purchasing, transportation price discounts. *Oper. Res.* 41(1):48–59.
- Engvall S, Göthe-Lundgren M, Värbrand P (2004) The heterogeneous vehicle-routing game. *Transportation Sci.* 38(1):71–85.
- Farvolden JM, Powell WB (1994) Subgradient methods for the service network design problem. *Transportation Sci.* 28(3):256–272.
- Ferrell W, Ellis K, Kaminsky P, Rainwater C (2020) Horizontal collaboration: Opportunities for improved logistics planning. *Internat. J. Production Res.* 58(14):4267–4284.
- Friesen DK, Langston MA (1986) Variable sized bin packing. *SIAM J. Comput.* 15(1):222–230.
- Gansterer M, Hartl RF (2018) Collaborative vehicle routing: A survey. *Eur. J. Oper. Res.* 268(1):1–12.
- Gansterer M, Hartl RF (2020) Shared resources in collaborative vehicle routing. *TOP* 28:1–20.
- Gendron B, Gouveia L (2017) Reformulations by discretization for piecewise linear integer multicommodity network flow problems. *Transportation Sci.* 51(2):629–649.
- Gillies D (1959) Solutions to general non-zero sum games. Tucker A, Luce R, eds. *Contributions to the Theory of Games IV* (Princeton University Press, Princeton, NJ), 47–83.
- Göthe-Lundgren M, Jörnsten K, Värbrand P (1996) On the nucleus of the basic vehicle routing game. *Math. Programming* 72(1):83–100.
- Guajardo M, Rönnqvist M (2016) A review on cost allocation methods in collaborative transportation. *Internat. Transportation Oper. Res.* 23(3):371–392.
- Guajardo M, Rönnqvist M, Flisberg P, Frisk M (2018) Collaborative transportation with overlapping coalitions. *Eur. J. Oper. Res.* 271(1):238–249.
- Hezarkhani B, Slikker M, Van Woensel T (2016) A competitive solution for cooperative truckload delivery. *OR Spectrum* 38(1):51–80.
- Hezarkhani B, Slikker M, Van Woensel T (2019) Gain-sharing in urban consolidation centers. *Eur. J. Oper. Res.* 279(2):380–392.
- Houghtalen L, Ergun Ö, Sokol J (2011) Designing mechanisms for the management of carrier alliances. *Transportation Sci.* 45(4): 465–482.
- iGPS (2019) Cross docking as a supply chain strategy. Accessed June 10, 2020, <https://igps.net/blog/pallet-pooling/supply-chain-lowering-your-total-cost-of-business/2019/10/10/cross-docking-as-a-supply-chain-strategy/>.
- Inbound Logistics (2016) CaseStack, Inc.: Logistics planner profiles. Accessed March 10, 2019, <http://www.inboundlogistics.com/cms/planner/2016/casestack-inc/>.
- Kaewpuang R, Niyato D, Tan PS, Wang P (2017) Cooperative management in full-truckload and less-than-truckload vehicle system. *IEEE Transportation Vehicular Tech.* 66(7):5707–5722.
- Kimms A, Kozeletskyi I (2016) Core-based cost allocation in the cooperative traveling salesman problem. *Eur. J. Oper. Res.* 248(3): 910–916.
- Li Z, Bookbinder JH, Elhedhli S (2012) Optimal shipment decisions for an airfreight forwarder: Formulation and solution methods. *Transportation Res. Part C: Emerging Tech.* 21(1):17–30.
- Lojistic (2019) How to calculate less-than-truckload LTL freight rates. Accessed March 10, 2019, <https://www.lojistic.com/ltl-freight-rates>.
- Marinakis Y, Migdalas A, Pardalos PM (2008) Cost allocation in combinatorial optimization games. Pardalos PM, Migdalas A, Pitsoulis L, eds. *Pareto Optimality, Game Theory and Equilibria* (Springer, New York), 217–247.
- Markakis E, Saberi A (2005) On the core of the multicommodity flow game. *Decision Support Systems* 39(1):3–10.
- Nguyen C, Dessouky M, Toriello A (2014) Consolidation strategies for the delivery of perishable products. *Transportation Res. Part E: Logist. Transportation Rev.* 69:108–121.
- Osicka O, Guajardo M, van Oost T (2020) Cooperative game-theoretic features of cost sharing in location-routing. *Internat. Transportation Oper. Res.* 27(4):2157–2183.
- Özener OÖ, Ergun Ö (2008) Allocating costs in a collaborative transportation procurement network. *Transportation Sci.* 42(2): 146–165.
- Özener OÖ, Ergun Ö, Savelsbergh M (2013) Allocating cost of service to customers in inventory routing. *Oper. Res.* 61(1):112–125.
- Pan S, Trentesaux D, Ballot E, Huang GQ (2019) Horizontal collaborative transport: Survey of solutions and practical implementation issues. *Internat. J. Production Res.* 57(15–16):5340–5361.
- Poljak BT (1987) *Introduction to Optimization*, vol. 1 (Optimization Software, Inc., Publications Division, New York).
- Ramaekers K, Verdonck L, Caris A, Meers D, Macharis C (2017) Allocating collaborative costs in multimodal barge networks for freight bundling. *J. Transportation Geography* 65:56–69.
- Robinson A (2014) 5 best transportation management optimization practices utilizing strategic freight shipping. Accessed March 10, 2018, <https://cerasis.com/2014/09/24/transportation-management-optimization/>.
- Robinson CH (2016) Assessing the 5 biggest LTL savings opportunities. Accessed March 10, 2018, <https://www.chrobinson.com/en-US/Resources/White-Papers/>.
- Russell RM, Krajewski LJ (1992) Coordinated replenishments from a common supplier. *Decision Sci.* 23(3):610–632.
- Shapley L (1971) Cores of convex games. *Internat. J. Game Theory* 1(1):11–26.

- Shapley L, Shubik M (1966) Quasi-cores in a monetary economy with nonconvex preferences. *Econometrica* 34(4):805–827.
- Terry L (2015) Collaborative distribution: Taking off the training wheels. Accessed March 10, 2019, <http://www.inboundlogistics.com/cms/article/collaborative-distribution-taking-off-the-training-wheels/>.
- Tinoco SVP, Creemers S, Boute RN (2017) Collaborative shipping under different cost-sharing agreements. *Eur. J. Oper. Res.* 263(3):827–837.
- Unishippers Global Logistics (2019) 5 factors that determine LTL shipping rates. Accessed March 10, 2019, <https://www.unishippers.com/ublog/5-factors-determine-ltl-freight-shipping-rates/>.
- van Zon M, Spliet R, van den Heuvel W (2021) The joint network vehicle routing game. *Transportation Sci.* 55(1):179–195.
- Vanovermeire C, Sörensen K (2014a) Integration of the cost allocation in the optimization of collaborative bundling. *Transportation Res. Part E: Logist. Transportation Rev.* 72(2):125–143.
- Vanovermeire C, Sörensen K (2014b) Measuring and rewarding flexibility in collaborative distribution, including two-partner coalitions. *Eur. J. Oper. Res.* 239(1):157–165.
- Vanovermeire C, Sörensen K, Van Breedam A, Vannieuwenhuyse B, Verstrepen S (2014) Horizontal logistics collaboration: Decreasing costs through flexibility and an adequate cost allocation strategy. *Internat. J. Logist. Res. Appl.* 17(4):339–355.
- Wolsey LA (1998) *Integer Programming*, vol. 42 (John Wiley & Sons, Inc., New York).
- Yilmaz O, Savaseneril S (2012) Collaboration among small shippers in a transportation market. *Eur. J. Oper. Res.* 218(2):408–415.
- Zhang W, Uhan NA, Dessouky M, Toriello A (2018) Moulin mechanism design for freight consolidation. *Transportation Res. Part B: Methodological* 116:141–162.
- Zheng J, Gao Z, Yang D, Sun Z (2015) Network design and capacity exchange for liner alliances with fixed and variable container demands. *Transportation Sci.* 49(4):886–899.